Stellar Atmospheres

Course B of two · Physical processes, transition rates, LTE and non-LTE, model atmospheres, with worked examples and exam questions & answers

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Final Examination — Course B: Stellar Atmospheres (with complete solutions)

Part A — Short answers (2 marks each)
Part B — Problems
Part C — Essay (choose one, 10 marks)
Part D — Regimes and opacity (4 marks each)

Formula Sheet — Course B: Stellar Atmospheres

LTE (Lecture SA1)
Continuous opacity (Lecture SA2)
Transition rates (Lecture SA3)
Line absorption and broadening (Lecture SA4)
Gray LTE atmosphere (Lecture SA5)
Lines (Lecture SA6)
TE, LTE, NLTE (Lecture SA7)
Stellar atmospheres (Lecture SA8)

Lectures SA0–SA8, final exam (SA9) and formula sheet (SA10). Prerequisite: Course A, *Radiative Transfer* (RT1–RT10).

Based on: D. Mihalas, *Stellar Atmospheres* (1978) · G. W. Collins II, *Fundamentals of Stellar Astrophysics*, Part II · D. F. Gray, *The Observation and Analysis of Stellar Photospheres* · S. Chandrasekhar, *Radiative Transfer* (1950) · Hubeny & Mihalas, *Theory of Stellar Atmospheres* (2014)

Code: figures and numbers come from `scripts/lte_solvers.py`, `scripts/sa_physics.py` and `scripts/worked_examples.py`.

Lecture SA0 — Course Overview: Stellar Atmospheres

> **Course B of two.** Course A, *Radiative Transfer* (lectures RT1–RT10), provides the transfer theory. This course applies it to stars and adds the **physical processes and transition rates** that determine opacities, emissivities and level populations, following the structure of Mihalas, *Stellar Atmospheres* (2nd ed., 1978): radiation field and transfer → absorption cross-sections → statistical equilibrium and rates → line broadening → LTE and NLTE model atmospheres and line formation. **Main sources:** D. Mihalas, *Stellar Atmospheres* (1978); G. W. Collins II, *Fundamentals of Stellar Astrophysics* (2003), Part II; D. F. Gray, *The Observation and Analysis of Stellar Photospheres* (2005); I. Hubeny & D. Mihalas, *Theory of Stellar Atmospheres* (2014); Rybicki & Lightman (1979).

What is a stellar atmosphere?

The **atmosphere** is the thin outer layer of a star from which photons escape into space: from optical depth $\tau \approx 10$–100 (where the gas is still opaque) out to $\tau \ll 1$. It is less than 0.1 % of the radius of a Sun-like star, yet it produces the entire observed spectrum. Collins (§9.1) stresses that it is defined by *the failure of the assumptions that work in the interior*: near the surface the photons begin to escape, the radiation field stops being Planckian, and the physics of the transfer of radiation becomes global instead of local.

The questions this course answers

1. **What is the state of the gas?** Temperature, pressure, ionisation and excitation (SA1, SA7).
2. **How does matter absorb and emit?** Bound–bound, bound–free, free–free and scattering processes: the opacity and emissivity (SA2, SA4).
3. **What sets the level populations?** Collisions versus radiation: transition rates and statistical equilibrium (SA3, SA7).
4. **What structure results?** Radiative equilibrium plus hydrostatic equilibrium: the model atmosphere (SA5, SA8).
5. **What spectrum emerges?** Continuum, limb darkening and spectral lines (SA5, SA6).

Prerequisites from Course A

You need	From
specific intensity, flux, moments	RT2
Planck function	RT3
α_ν, j_ν, Einstein coefficients, scattering source function	RT4
formal solution; MIT two-layer star	RT5
plane-parallel RTE, Eddington–Barbier	RT7
moments, Eddington approximation, Rosseland mean, thermalisation	RT8

Course map

#	Lecture	Key content
SA0	Overview (this page)	what an atmosphere is; course plan
SA1	Thermodynamic state: LTE, Saha–Boltzmann, Kirchhoff	populations in LTE, two-level atom
SA2	**Physical processes I: continuous opacity**	hydrogenic bound–free and free–free, H^-, scattering, opacity tables
SA3	**Physical processes II: transition rates and statistical equilibrium**	radiative and collisional rates, detailed balance, Milne relations, coronal and nebular limits
SA4	**Line absorption and broadening**	oscillator strengths, natural, Doppler, Stark, van der Waals broadening, Voigt profile, rotation
SA5	The gray LTE atmosphere: complete solution	$T^4 = \frac{3}{4}T_{\rm eff}^4(\tau + q)$, Hopf function, limb darkening, spectrum
SA6	LTE spectral lines	Milne–Eddington, curve of growth, abundances
SA7	TE, LTE and non-LTE	definitions with examples, rate equations, NLTE mechanisms and methods
SA8	Model atmospheres and the solar atmosphere	structure equations, blanketing, convection, solar layers, grids, diagnostics
SA9	Final exam with full solutions	
SA10	Formula sheet	

Collins' five assumptions of "normal" stellar atmospheres

1. **LTE**: all material properties follow from the local T and ρ.
2. The structure is set by the **continuum** opacity.
3. **Plane-parallel** geometry: thickness much smaller than the radius.
4. **Constant gravity** g, with hydrostatic equilibrium $dP/dz = -\rho g$.
5. **Radiative equilibrium**, with no energy sources: $F = \sigma T_{\rm eff}^4$ = constant.

Under these assumptions only $T_{\rm eff}$, $\log g$ and the chemical composition specify the atmosphere. Lectures SA5–SA6 work entirely inside these assumptions. SA7–SA8 show where each one fails and what replaces it.

Lecture SA1 — Thermodynamic Equilibrium and Local Thermodynamic Equilibrium (LTE)

> **Sources:** Collins §9.1a–d (breakdown of STE, assumption of LTE, the five assumptions of normal stellar atmospheres), §9.2c (Kirchhoff's law $j_\nu = \kappa_\nu B_\nu$, eq. 9.2.22), §9.2d ($S_\nu = B_\nu$, eq. 9.2.28); MIT 8.901 §7.3 (Kirchhoff's law for thick and thin lumps); Hu Set 1; Carroll & Ostlie Ch. 8 (Saha–Boltzmann example).

Learning goals

1. State what is in equilibrium in TE and in LTE.
2. Derive Kirchhoff's law $j_\nu = \alpha_\nu B_\nu(T)$ and the LTE result $S_\nu = B_\nu(T)$.
3. Use the Boltzmann and Saha equations; explain the strength of the Balmer lines along the spectral sequence.
4. Derive the two-level-atom source function and the condition for LTE to be valid.

Key concepts in plain words

Equilibrium means "the most probable arrangement". When particles collide very often, energy is shared out in the most probable way, and every distribution (speeds, excitation, ionisation, photons) is fixed by a single number: the temperature. That is thermodynamic equilibrium (TE). It is like a crowded room where everyone constantly exchanges coins: after a while the distribution of coins no longer depends on how it started.

Why the radiation is the first to leave equilibrium. In a stellar atmosphere, particles (atoms, ions, electrons) travel only a tiny distance between collisions, but photons can travel much farther, especially near the surface where the gas is thin. Photons near the top can simply escape into space, and no photons come back from space to replace them. So the radiation field becomes **non-equilibrium** (anisotropic, deficient in outward-escaping light) long before the particles notice anything. Collins calls this the key asymmetry between photons and massive particles.

LTE — a local thermostat for matter. If collisions are still frequent enough, each small volume of gas keeps its particles in equilibrium at the **local** temperature: Maxwellian speeds, Boltzmann excitation, Saha ionisation. That is LTE. The radiation passing through can be anything it likes. In LTE the gas emits exactly as much, per unit absorption, as a black body at the local temperature would (Kirchhoff's law), so $S_\nu = B_\nu(T)$.

Kirchhoff's law in words. Good absorbers are good emitters. A material that absorbs strongly at some frequency also emits strongly at that frequency, and the ratio of emission to absorption is the same universal function $B_\nu(T)$ for every material in equilibrium. This is why a thick, isothermal piece of anything glows as a black body.

Boltzmann — a tax on climbing the energy ladder. Atomic levels are like steps on a ladder. Occupying a higher step costs energy ΔE, and in equilibrium the fraction of atoms on that step is reduced by the factor $e^{-\Delta E/kT}$ (a "tax" that is heavy when kT is small), multiplied by the number of available states g.

Saha — energy cost versus freedom. Ionising an atom costs a large energy χ, which discourages ionisation. But a freed electron gains an enormous number of possible states (positions and velocities), which favours ionisation. Saha's equation balances these two effects. Because the number of free-electron states is huge, hydrogen becomes ionised at

about 10 000 K, even though χ/k is 158 000 K. Low density favours ionisation too (an electron that is set free rarely meets an ion to recombine with).

Who controls the atoms: collisions or radiation? An excited atom can be de-excited by a collision (energy goes to the gas, pushing toward LTE) or by emitting a photon (energy goes to the radiation, which may escape). If collisions are much faster ($C \gg A$, high density), the populations follow the local temperature: LTE. If radiation wins (low density, strong lines), the populations follow the radiation field: NLTE. The **critical density** $n_{\text{crit}} = A/q$ marks the change-over.

1.1 Strict (full) thermodynamic equilibrium

In TE at temperature T, **every** distribution is the equilibrium one:

Distribution	Law
particle velocities	Maxwell: $f(v) \propto v^2 e^{-mv^2/2kT}$
excitation	Boltzmann: $\frac{n_j}{n_i} = \frac{g_j}{g_i} e^{-(E_j - E_i)/kT}$
ionisation	Saha (below)
radiation	Planck: $I_\nu = B_\nu(T)$, isotropic

Deep in a star this holds extremely well. The photon mean free path ($\sim$ cm) is tiny compared with the temperature scale height ($\sim R_\odot \sim 10^{11}$ cm), so every photon sees an isothermal enclosure.

1.2 Why TE fails near the surface

Collins (§9.1a) identifies the key asymmetry: **photons have much longer mean free paths than particles**. Approaching the surface:

1. Photons begin to **escape** into space, so I_ν becomes anisotropic and departs from B_ν. The radiation field “feels” the boundary first.
2. A photon at a given point was emitted in regions with different T; the radiation field is **non-local**.
3. Particles, with short mean free paths, still collide often enough to keep Maxwell, Boltzmann and Saha distributions at the **local** T.

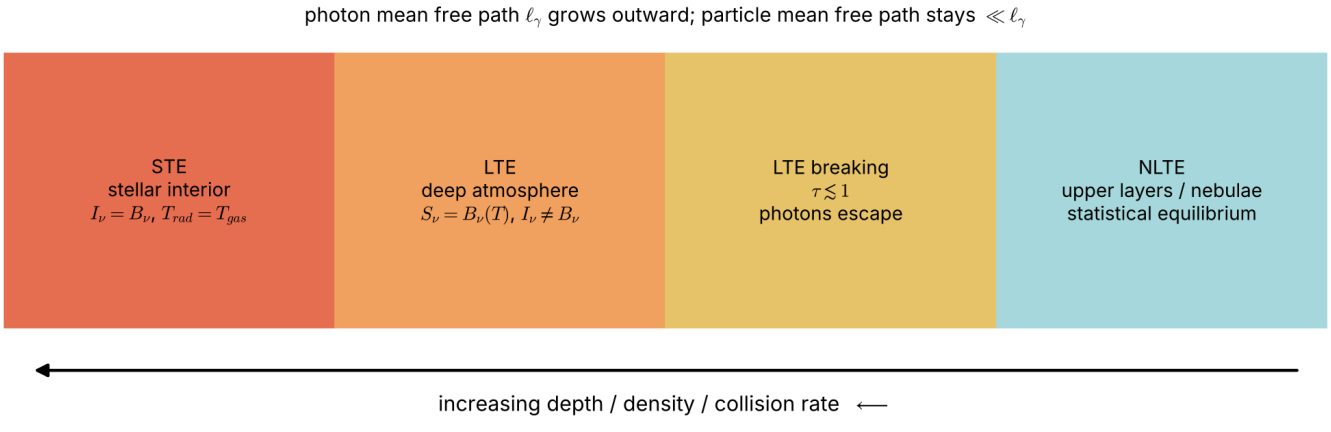


Figure 1.1 — From STE in the interior, through LTE in the deep atmosphere, to NLTE in the upper layers. LTE fails first for strong lines and at low density.

1.3 Local thermodynamic equilibrium

> **Definition.** In **LTE** the *matter* at each point is in thermodynamic equilibrium at the local temperature $T(\mathbf{r})$: velocities are Maxwellian, populations follow Boltzmann and Saha. The radiation field I_ν is **not** required to be Planckian.

Kirchhoff's law

In a closed cavity in TE, $I_\nu = B_\nu$ is homogeneous and isotropic, so $dI_\nu/ds = 0$. The RTE then gives $0 = -\alpha_\nu B_\nu + j_\nu$:

$$\boxed{j_\nu = \alpha_\nu B_\nu(T)} \qquad \text{(Kirchhoff's law)}$$

Both j_ν and α_ν depend only on the **state of the matter** (populations and velocity distribution). In LTE that state is the TE state at the local T, so Kirchhoff's law still holds, even though $I_\nu \neq B_\nu$. Hence:

$$\boxed{S_\nu^{\rm LTE} = \frac{j_\nu}{\alpha_\nu} = B_\nu(T)}$$

MIT 8.901 (§7.3) stresses the other half of this statement: $S_\nu = B_\nu$ does **not** mean the object emits a black-body spectrum. An optically thick lump emits $B_\nu(T)$, but an optically thin lump at the same T emits only $\tau_\nu B_\nu(T)$, which is bright only at the frequencies where α_ν is large (see Lecture RT5, §5.4).

Collins (eq. 9.2.28) notes that $S_\nu = B_\nu$ is **sufficient** for LTE but not necessary in general: with electron scattering present, $S_\nu = \varepsilon B_\nu + (1-\varepsilon) J_\nu$, yet the populations may still be in LTE.

LTE written with Einstein coefficients

With Boltzmann populations, $g_1 n_2 / (g_2 n_1) = e^{-h\nu/kT}$, the formulas of Lecture RT4 give

$$\alpha_\nu = \frac{h\nu}{4\pi} n_1 B_{12} \phi(\nu) \left(1 - e^{-h\nu/kT}\right), \qquad \frac{j_\nu}{\alpha_\nu} = \frac{2h\nu^3}{c^2} \frac{1}{e^{h\nu/kT} - 1} = B_\nu(T). \;\checkmark$$

The factor $\left(1 - e^{-h\nu/kT}\right)$ is the **LTE stimulated-emission correction**; it multiplies every LTE opacity.

What LTE does and does not say

True in LTE	Not implied by LTE
$S_\nu = B_\nu(T_{\rm local})$	$I_\nu = B_\nu$
populations from Saha–Boltzmann at $T_{\rm local}$	$J_\nu = B_\nu$ (it fails near the surface)
opacity and emissivity are functions of (T, ρ) only	isotropic radiation
once $T(\tau)$ is known, the formal solution gives I_ν directly	zero flux

> **See also Lecture SA7**, which gives precise side-by-side definitions of **TE, LTE and NLTE**, with astrophysical examples of each (stellar core, early Universe; photospheric continuum, Earth's troposphere; resonance-line cores, chromospheres, nebulae, masers) and a quantitative criterion for deciding which applies.

1.4 Boltzmann and Saha equations

Boltzmann (excitation within one ionisation stage):

$$\frac{N_j}{N_i} = \frac{g_j}{g_i}\, e^{-(E_j - E_i)/kT}, \qquad \frac{N_j}{N} = \frac{g_j\, e^{-E_j/kT}}{U(T)}, \quad U = \sum_i g_i e^{-E_i/kT}.$$

For hydrogen, $g_n = 2n^2$ and $E_n = -13.6\ \mathrm{eV}/n^2$.

Saha (ionisation $r \to r+1$):

$$\boxed{\frac{N_{r+1}\, n_e}{N_r} = \frac{2U_{r+1}}{U_r}\left(\frac{2\pi m_e kT}{h^2}\right)^{3/2} e^{-\chi_r/kT}}$$

with $(2\pi m_e k/h^2)^{3/2} = 2.415 \times 10^{21}\ \mathrm{m^{-3}\,K^{-3/2}}$. Using $P_e = n_e kT$:

$$\frac{N_{r+1}}{N_r} = \frac{2kT\, U_{r+1}}{P_e\, U_r}\left(\frac{2\pi m_e kT}{h^2}\right)^{3/2} e^{-\chi_r/kT}.$$

The factor 2 is the spin degeneracy of the free electron. The $T^{3/2}$ counts the free-electron phase space, and that large number is why gases ionise at $kT \ll \chi$.

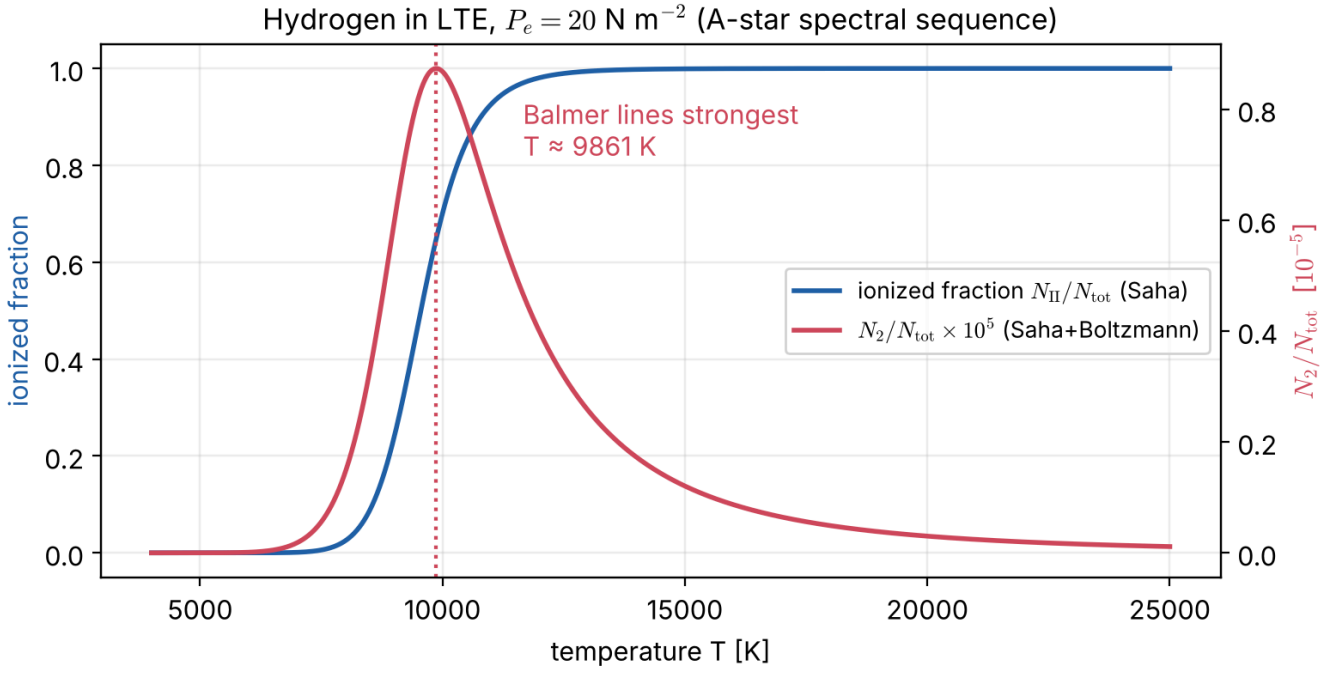


Figure 1.2 — Hydrogen in LTE at $P_e = 20$ N m^{-2}. The $n = 2$ population, which produces the Balmer lines, rises with T (Boltzmann) and then collapses as H ionises (Saha). Its peak at $T \approx 9900$ K explains why Balmer lines are strongest in A0 stars.

1.5 When is LTE valid? The two-level atom

Consider the line $1 \leftrightarrow 2$ with collisional rates C_{12}, C_{21} (per atom per second; $C_{21} = n_e q_{21}$). Detailed balance for collisions gives $C_{12}/C_{21} = (g_2/g_1)e^{-h\nu/kT}$. Statistical equilibrium:

$$n_1 \left(B_{12}\bar{J} + C_{12}\right) = n_2 \left(A_{21} + B_{21}\bar{J} + C_{21}\right).$$

Insert this into $S = (2h\nu^3/c^2)/[g_2 n_1/(g_1 n_2) - 1]$ and use the Einstein relations. After some algebra (Exam Q4):

$$\boxed{S_\nu^{\text{line}} = (1-\varepsilon)\bar{J} + \varepsilon B_\nu(T), \qquad \varepsilon = \frac{\varepsilon'}{1+\varepsilon'}, \quad \varepsilon' = \frac{C_{21}}{A_{21}}\left(1 - e^{-h\nu/kT}\right)}$$

- **Collisions dominate** ($C_{21} \gg A_{21}$, high density): $\varepsilon \to 1$, $S \to B$: **LTE**.
- **Radiation dominates** (low density): $\varepsilon \ll 1$, $S \approx \bar{J}$: scattering-like, **NLTE**.

The **critical density** $n_{\text{crit}} = A_{21}/q_{21}$ separates the two. The line source function has exactly the form found for continuum scattering in Lecture RT4, so the methods of Lecture RT8 apply.

Collins' five assumptions of "normal" stellar atmospheres (§9.1d)

1. LTE prevails: all material properties follow from the local T and ρ.
2. The atmospheric structure is set by the continuum opacity.
3. The geometry is a plane-parallel slab (thickness $\ll R$).
4. Gravity g is constant through the atmosphere: $dP/dz = -g\rho$.
5. All energy is carried by radiation, with no sources: $F = \sigma T_{\text{eff}}^4$ = constant.

Then **only T_{eff}, g and the composition** specify the atmosphere. This is the basis of Lectures RT7–RT8 and SA5–SA6.

Worked example 1.1 — Why are the Balmer lines weak in the Sun?

At $T = 5772$ K, compute N_2/N_1 for hydrogen. $kT = 0.4974$ eV, so $N_2/N_1 = (8/2)e^{-10.20/0.4974} = 4e^{-20.51} = 5.0 \times 10^{-9}$. Only five atoms in a billion can absorb a Balmer photon. The Balmer lines are still visible because hydrogen is so abundant. At 9900 K: $N_2/N_1 = 4e^{-11.96} = 2.6 \times 10^{-5}$, about 5000 times larger.

Worked example 1.2 — Saha ionisation at 9900 K

Hydrogen at $T = 9900$ K, $P_e = 20$ Pa. Find $N_{\text{II}}/N_{\text{I}}$. Take $U_{\text{I}} = 2$, $U_{\text{II}} = 1$.

1. $n_e = P_e/kT = 20/(1.381 \times 10^{-23} \times 9900) = 1.46 \times 10^{20}$ m^{-3}.
2. $(2\pi m_e kT/h^2)^{3/2} = 2.415 \times 10^{21} \times 9900^{3/2} = 2.38 \times 10^{27}$ m^{-3}.
3. $e^{-13.6/0.8531} = e^{-15.94} = 1.19 \times 10^{-7}$.
4. $N_{\text{II}}/N_{\text{I}} = \dfrac{2}{2} \times \dfrac{2.38 \times 10^{27} \times 1.19 \times 10^{-7}}{1.46 \times 10^{20}} = 1.94$.

The ionised fraction is $1.94/2.94 = 0.66$. Then $N_2/N_{\text{tot}} = (2.6 \times 10^{-5}/1.000026) \times (1/2.94) \approx 8.7 \times 10^{-6}$, which is the peak in Figure 1.2.

Worked example 1.3 — Is the H I 21-cm line in LTE in the ISM?

$A_{21} = 2.9 \times 10^{-15}\ s^{-1}$, and the collisional de-excitation rate coefficient is $q_{21} \approx 10^{-16}\ m^3\ s^{-1}$ (H–H collisions at 100 K). Is the line in LTE at $n_{\rm H} = 10^6\ m^{-3}$ (1 cm^{-3})?

$n_{\rm crit} = A_{21}/q_{21} \approx 29\ \mathrm{m}^{-3} \ll 10^6\ \mathrm{m}^{-3}$. Collisions dominate, so the spin temperature is close to the kinetic temperature (the populations are thermalised). By contrast, optical forbidden lines such as [O III] have $A \sim 10^{-2}\ \mathrm{s}^{-1}$ and $n_{\rm crit} \sim 10^{11}$–$10^{12}\ \mathrm{m}^{-3}$, above nebular densities, so they are strongly NLTE.

Common misconceptions

1. **"LTE means the radiation is Planckian."** LTE concerns matter; I_ν can be far from B_ν (e.g. no incoming light at the surface).
2. **"Hydrogen ionises when $kT \approx 13.6$ eV."** It ionises at $kT \approx 0.9$ eV, because of the huge number of free-electron states (Saha).
3. **"If a gas is dense it is in LTE for every line."** Strong lines can still be NLTE in the upper layers, because photons escape there.
4. **"Excitation temperature always equals kinetic temperature."** Only in LTE. In NLTE they can differ greatly (e.g. masers have negative $T_{\rm ex}$).

Lecture summary

- TE: one temperature for matter and radiation, $I_\nu = B_\nu$. LTE: matter at local T, radiation free.
- Kirchhoff: $j_\nu = \alpha_\nu B_\nu(T) \Rightarrow S_\nu = B_\nu(T)$ in LTE.
- Boltzmann (excitation) and Saha (ionisation) give populations; together they explain the spectral sequence.
- Two-level atom: $S = (1-\varepsilon)\bar{J} + \varepsilon B$. LTE when collisions dominate ($\varepsilon \to 1$).

Exam questions and answers — Lecture SA1

Q1. Define LTE precisely. Explain why LTE can hold even though the radiation field near a stellar surface is very far from a black body.

> **A1.** LTE: at each point the matter (velocity, excitation and ionisation distributions) is in TE at the local temperature $T(\mathbf{r})$. It can hold because particle–particle collisions, which set the material distributions, are frequent (short particle mean free path). Photons have long mean free paths and escape, so $I_\nu \neq B_\nu$, but as long as collisional rates dominate the radiative rates in setting the populations, the matter does not "notice" the anomalous radiation field.

Q2. Derive Kirchhoff's law and show that it implies $S_\nu = B_\nu$ in LTE.

> **A2.** In a TE cavity, $I_\nu = B_\nu$ is uniform and isotropic, so $dI_\nu/ds = 0 = j_\nu - \alpha_\nu B_\nu$, giving $j_\nu = \alpha_\nu B_\nu(T)$. Because j_ν and α_ν are properties of the matter only, the relation holds whenever the matter is in its

TE state at the local T, i.e. in LTE. Then $S_\nu = j_\nu/\alpha_\nu = B_\nu(T)$.

Q3. Explain, with Saha and Boltzmann, why Balmer lines peak in strength for A0 stars ($T \approx 10^4$ K) and are weaker in both cooler and hotter stars.

A3. The Balmer opacity $\propto N_2$. In cool stars, $N_2/N_1 \propto e^{-10.2\,\mathrm{eV}/kT}$ is tiny (e.g. 5×10^{-9} at 5800 K). Above about 10^4 K, Saha ionisation removes neutral H: $N_\mathrm{I}/N_\mathrm{tot} \to 0$. The product N_2/N_tot therefore peaks near 9900 K (for $P_e = 20$ Pa).

Q4. For a two-level atom with collisions, derive $S = (1-\varepsilon)\bar{J} + \varepsilon B$. When does the line form in LTE?

A4. From statistical equilibrium, $\frac{n_1}{n_2} = \frac{A_{21}+B_{21}\bar{J}+C_{21}}{B_{12}\bar{J}+C_{12}}$. Then $\frac{g_2 n_1}{g_1 n_2} - 1 = \frac{A_{21}+C_{21}(1-e^{-h\nu/kT})}{B_{21}\bar{J}+C_{21}e^{-h\nu/kT}}$ (using $g_2B_{21} = g_1B_{12}$ and $g_2C_{21}e^{-h\nu/kT} = g_1C_{12}$). So $S = \frac{2h\nu^3}{c^2}\frac{B_{21}\bar{J}+C_{21}e^{-h\nu/kT}}{A_{21}+C_{21}(1-e^{-h\nu/kT})}$. With $A_{21} = (2h\nu^3/c^2)B_{21}$ and $\varepsilon' = C_{21}(1-e^{-h\nu/kT})/A_{21}$: $S = \frac{\bar{J}+\varepsilon' B_\nu}{1+\varepsilon'}$, because $\frac{2h\nu^3}{c^2}\frac{e^{-h\nu/kT}}{1-e^{-h\nu/kT}} = B_\nu$. Writing $\varepsilon = \varepsilon'/(1+\varepsilon')$ gives the result. LTE holds when $C_{21} \gg A_{21}$, i.e. $n_e \gg n_\mathrm{crit}$, but **also** only at depths $\gtrsim 1/\sqrt{\varepsilon}$ (Lecture RT8).

Lecture SA2 — Physical Processes I: Continuous Opacity

Sources: D. Mihalas, *Stellar Atmospheres* (1978), absorption cross-sections (bound–free, free–free, H^-, scattering); D. F. Gray, *The Observation and Analysis of Stellar Photospheres* (2005), Ch. 8 “The continuous absorption coefficient”; Rybicki & Lightman (1979), Chs. 5 (bremsstrahlung) and 10 (atomic structure); Wildt (1939) for H^-; Collins Ch. 9. **Prerequisites:** RT4 (absorption coefficient, cross-section, stimulated emission), SA1 (Saha–Boltzmann populations).

Learning goals

1. List the physical processes that make up the continuous opacity of a stellar atmosphere and say where each one dominates.
2. Use Kramers’ hydrogenic bound–free cross-section and the free–free formula with LTE populations.
3. Explain why H^- is the dominant opacity source in the Sun and why its strength depends on metallicity.
4. Explain the Balmer jump, and combine the processes into a total opacity and a mean opacity.

Key concepts in plain words

Opacity is a menu of processes. A photon in a stellar atmosphere can be stopped in four basic ways: (1) it lifts a bound electron to a higher bound level (*bound–bound*: spectral lines, SA4); (2) it frees a bound electron (*bound–free*: photo-ionisation); (3) a free electron absorbs it while passing an ion (*free–free*: inverse bremsstrahlung); (4) it is redirected without being destroyed (*scattering*). Processes 2–4 vary smoothly with frequency and make the **continuous opacity**, which sets the overall shape of the spectrum and the depth scale of the atmosphere.

Edges are the fingerprints of bound–free absorption. A photon can ionise an atom from level n only if its energy exceeds the binding energy of that level. Each level therefore contributes absorption that starts abruptly at a threshold frequency (an *edge*) and fades toward higher frequency roughly as ν^{-3}. For hydrogen the edges are at 91.2 nm (Lyman), 364.7 nm (Balmer) and 820.6 nm (Paschen).

The opacity depends on the populations. A process can absorb only if there are atoms in the right state. In LTE these populations come from Saha and Boltzmann (SA1), so the opacity changes dramatically with temperature: at 6000 K almost no hydrogen atoms are in level 2, while at 9000 K many more are.

H^-: a fragile ion with a big effect. A neutral hydrogen atom can hold a second electron, but only with 0.754 eV of binding. In the Sun only about one hydrogen atom in 10^8 is H^-. That is enough to dominate the visible and infrared opacity, because it can be ionised by any photon with $\lambda < 1.64$ µm, while ordinary hydrogen in cool gas is almost all in the ground state and absorbs only below 91 nm. The extra electrons come from easily ionised metals, so H^- opacity grows with metallicity.

2.1 The four kinds of continuous and line processes

Process	Photon is	Frequency dependence	Dominant in
bound–bound	absorbed (line)	narrow profiles (SA4)	lines in all stars
bound–free	absorbed (ionisation)	edges, $\propto \nu^{-3}$ above each edge	H I in A–B stars; He in O–B; metals in the UV; H^- in F–K
free–free	absorbed (inverse bremsstrahlung)	$\propto \nu^{-3}$ (optical), ν^{-2} (radio)	hot stars (IR, radio); H^- free–free in cool stars beyond 1.6 μm
electron scattering	scattered (Thomson)	gray	O stars, stellar interiors
Rayleigh scattering	scattered by atoms or molecules	$\propto \lambda^{-4}$	blue and UV of cool giants

2.2 Hydrogenic bound–free absorption

Kramers' cross-section for photo-ionisation from level n of a hydrogen-like ion of charge Z (semi-classical, corrected by a Gaunt factor $g_{\rm bf} \approx 1$):

$$\boxed{\sigma_{\rm bf}(n,\nu) = \frac{64\pi^4 m_e e^{10}}{3\sqrt{3}\,ch^6}\,\frac{Z^4}{n^5\nu^3}\,g_{\rm bf} = 2.815\times10^{29}\,\frac{Z^4 g_{\rm bf}}{n^5\nu^3}\ \text{cm}^2 \quad (\nu \ge \nu_n)}$$

(in SI: 2.815×10^{25} m^2 Hz^3; first formula in Gaussian units). The thresholds are $\nu_n = 3.288\times10^{15}\,Z^2/n^2$ Hz, i.e. $\lambda_n = 91.13\,n^2/Z^2$ nm.

Absorption coefficient in LTE. Sum over levels with Boltzmann populations, $N_n = N_{\rm H\,I}\,2n^2 e^{-E_n/kT}/U$, and include stimulated emission:

$$\alpha_\nu^{\rm bf} = \sum_n N_n\,\sigma_{\rm bf}(n,\nu)\left(1 - e^{-h\nu/kT}\right).$$

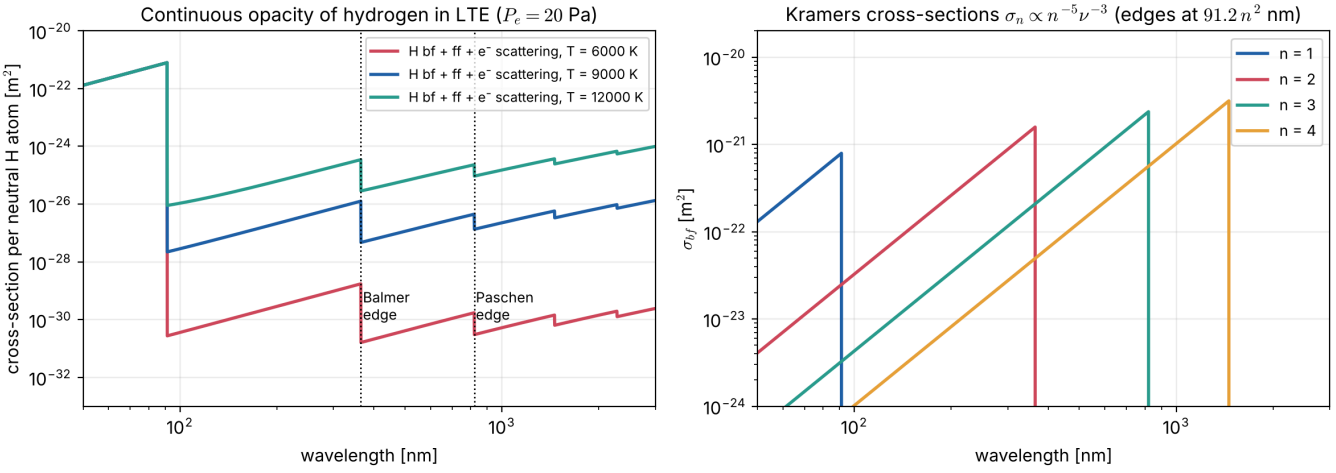


Figure 2.1 — Left: hydrogen continuous cross-section per neutral H atom (bound–free + free–free + electron scattering) in LTE at $P_e = 20$ Pa, computed with `scripts/sa_physics.py`. The jumps at the Lyman, Balmer and Paschen edges grow dramatically with temperature, because the excited-level populations follow Boltzmann. Right: Kramers cross-sections of the first four levels. Each level absorbs only blueward of its edge.

2.3 Free–free absorption

A free electron passing an ion can absorb a photon (the inverse of bremsstrahlung). For a hydrogen-like ion in LTE (Rybicki & Lightman eq. 5.18b, Gaussian units, n in cm^{-3}):

$$\alpha_\nu^{\rm ff} = 3.69\times10^{8}\,\frac{Z^2 n_e n_i}{T^{1/2}\nu^3}\,g_{\rm ff}\left(1-e^{-h\nu/kT}\right)\ \text{cm}^{-1}.$$

It rises toward low frequency, dominates the infrared and radio emission of hot ionised gas (H II regions, RT5), and in cool stars appears as H$^-$ free–free (an electron passing a *neutral* H atom).

2.4 The negative hydrogen ion H$^-$

- **Binding energy** 0.754 eV, so the bound–free threshold is at $\lambda = 1.64$ µm. The cross-section peaks near 850 nm at about 4×10^{-21} m^2 (4×10^{-17} cm^2).
- **Free–free** H$^-$ (electron + neutral H) dominates beyond about 1.6 µm. The minimum of the total H$^-$ opacity near 1.6 µm makes the solar continuum at that wavelength come from the deepest visible layers.
- **Abundance** from the Saha equation (H$^-$ has $U=1$, H I has $U=2$, one electron added):

$$\frac{n(\mathrm{H^-})}{n(\mathrm{H\,I})\,n_e} = \frac{1}{4}\left(\frac{h^2}{2\pi m_e kT}\right)^{3/2} e^{+0.754\,\mathrm{eV}/kT}.$$

- **Electron donors.** In the solar photosphere hydrogen is nearly neutral; the electrons come mainly from easily ionised metals (Na, Mg, Al, Si, Ca, Fe; ionisation energies 5–8 eV). The H$^-$ opacity is therefore roughly proportional to the metal abundance: metal-poor stars are more transparent, and we see deeper and hotter layers in them.
- **History:** Wildt (1939) identified H$^-$ as the main solar opacity; Chandrasekhar computed its cross-sections in the 1940s.

2.5 Scattering and other sources

- **Electron (Thomson) scattering**, $\sigma_T = 6.652\times10^{-29}$ m^2, gray. $\kappa_{\rm es} = 0.020(1+X)$ m^2 kg^{-1} for fully ionised gas. It dominates in O stars and enters the source function as $S = \varepsilon B + (1-\varepsilon)J$ (RT8).
- **Rayleigh scattering** by H I and H_2, $\propto \lambda^{-4}$ far from resonance. Important in the blue and UV of cool giants.
- **He I and He II bound–free** (edges at 50.4 and 22.8 nm, plus excited-level edges) in hot stars; **metal bound–free** edges (Mg, Si, Fe, Al, C) in the UV of solar-type stars.
- **Molecules** (TiO, H_2O, CO, CH_4) and **dust** in M, L and T dwarfs.

2.6 Total and mean opacity

$$\kappa_\nu^{\rm tot}\rho = \sum_{\rm species}\left(\alpha_\nu^{\rm bf}+\alpha_\nu^{\rm ff}\right)\left(1-e^{-h\nu/kT}\right) + \alpha_\nu^{\rm es} + \alpha_\nu^{\rm Ray} + \alpha_\nu^{\rm lines}$$

(the stimulated-emission factor applies to true absorption only, not to scattering).

- Structure calculations need mean opacities: the **Rosseland mean** (harmonic, for the flux, RT8) and the **Planck mean** (arithmetic, for emission and absorption balance).
- **Opacity tables:** OPAL (Iglesias & Rogers 1996), the Opacity Project (Seaton et al. 1994), and low-temperature molecular and dust tables (e.g. Ferguson et al. 2005).
- In stellar interiors the bound–free and free–free opacities combine into **Kramers' law**, $\bar{\kappa} \propto \rho T^{-3.5}$.

Worked example 2.1 — The size of a bound–free cross-section

Evaluate Kramers' formula at the Lyman and Balmer edges of hydrogen ($g_{\rm bf} = 1$).

Lyman ($n = 1$, $\nu_1 = 3.288 \times 10^{15}$ Hz): $\sigma = 2.815 \times 10^{25}/(3.288 \times 10^{15})^3 = 7.9 \times 10^{-22}$ m^2 (the exact quantum value, with $g_{\rm bf} = 0.80$, is 6.3×10^{-22} m^2). Balmer ($n = 2$, $\nu_2 = \nu_1/4$): $\sigma = 2.815 \times 10^{25}/(2^5(\nu_1/4)^3) = 7.9 \times 10^{-22} \times 64/32 = 1.6 \times 10^{-21}$ m^2. **Note:** the edge cross-section *grows* with n (as n), because the ν^{-3} factor wins over n^{-5}. Excited levels are excellent absorbers, if they are populated.

Worked example 2.2 — Why H^- dominates the solar opacity at 500 nm

At $T = 6000$ K and $n_e = 10^{19}$ m^{-3}, compare the H^- bound–free cross-section per H atom with the hydrogen $n = 3$ (Paschen) bound–free cross-section per H atom at 500 nm.

1. **H^-:** $n(\mathrm{H^-})/n(\mathrm{H\,I}) = n_e / \left[4\,(2\pi m_e kT/h^2)^{3/2} e^{-0.754/kT}\right]$. Here $(2\pi m_e kT/h^2)^{3/2} = 1.12 \times 10^{27}$ m^{-3} and $e^{-0.754/0.517} = 0.233$, so the ratio is $10^{19}/(4 \times 1.12 \times 10^{27} \times 0.233) = 9.6 \times 10^{-9}$. Times $\sigma \approx 4 \times 10^{-21}$ m^2: 3.8×10^{-29} m^2 per H atom.
2. **H I, $n = 3$:** $N_3/N_{\rm H\,I} = 9e^{-12.09/0.517} = 6.3 \times 10^{-10}$; $\sigma_3(500\,\mathrm{nm}) = 5.4 \times 10^{-22}$ m^2. Product: 3.4×10^{-31} m^2 per H atom.
3. **Ratio** ≈ 110. A tiny fraction of fragile negative ions out-absorbs the excited neutral hydrogen by two orders of magnitude.

Worked example 2.3 — The Balmer jump across the spectral sequence

From the computation behind Figure 2.1 ($P_e = 20$ Pa), the hydrogen opacity just blueward of the Balmer edge exceeds that just redward of it by factors of 106 (6000 K), 26 (9000 K) and 12 (12 000 K). Why is the observed Balmer jump largest in late B and A stars, not in the Sun?

In the Sun hydrogen bound–free is not the main opacity: H^- is about 100 times larger (Worked example 2.2), and it has no edge at 364.7 nm. The hydrogen jump is diluted and the observed solar Balmer jump is small. Near 9000–10 000 K hydrogen dominates the continuum, so its jump appears fully in the spectrum: blueward of the edge the opacity is much higher, we see higher and cooler layers (Eddington–Barbier), and the flux drops. At higher T the jump shrinks (12 at 12 000 K) as electron scattering and ionisation take over. The Balmer jump therefore measures temperature in B stars and gravity (through P_e) in A and F stars.

Common misconceptions

1. **"The most abundant atom gives the most opacity."** What matters is how many particles are in an absorbing state at the relevant frequency. A 10^{-8} fraction of H^- beats neutral hydrogen in the Sun.

2. **“Bound–free absorption occurs only at the edge.”** It occurs at all frequencies *above* the edge, falling roughly as ν^{-3}.
3. **“Electron scattering is absorption.”** It redirects photons without destroying them; it adds to the extinction but enters the source function as J, not B.
4. **“Metals are too rare to matter for the continuum.”** They supply the electrons that make $\mathrm{H^-}$, and their bound–free edges dominate the UV of cool stars.

Lecture summary

- Continuous opacity = bound–free + free–free + scattering (plus $\mathrm{H^-}$, molecules, dust); lines are added separately (SA4).
- Kramers: $\sigma_{\rm bf} = 2.815\times10^{29}\,Z^4 g/(n^5\nu^3)$ cm^2 above $\nu_n = 3.29\times10^{15}\,Z^2/n^2$ Hz.
- Free–free: $\alpha_{\rm ff} \propto Z^2 n_e n_i T^{-1/2}\nu^{-3}$.
- $\mathrm{H^-}$ (0.754 eV, threshold 1.64 μm) dominates F–K photospheres; its strength scales with n_e, i.e. with metallicity.
- Edges produce the Balmer and Paschen jumps; the opacity changes enormously with T through Saha–Boltzmann.

Exam questions and answers — Lecture SA2

Q1. Write Kramers’ bound–free cross-section and show that, at the edge, the cross-section of level n is proportional to n.

> **A1.** $\sigma_{\rm bf}(n,\nu) = 2.815\times10^{29}\,Z^4 g_{\rm bf}/(n^5\nu^3)$ cm^2 for $\nu \geq \nu_n = \nu_1/n^2$. At the edge, $\nu_n^3 = \nu_1^3/n^6$, so $\sigma(n,\nu_n) = 2.815\times10^{29}\,Z^4 g\, n^6/(n^5\nu_1^3) \propto n$.

Q2. Explain why $\mathrm{H^-}$ is the main continuous opacity in the solar photosphere, and why metal-poor stars are more transparent.

> **A2.** In the Sun hydrogen is mostly neutral and in the ground state, which absorbs only below 91 nm. Excited levels are very weakly populated ($N_3/N_1 \sim 10^{-9}$). $\mathrm{H^-}$, though rare ($\sim 10^{-8}$ of H), can be ionised by any photon below 1.64 μm with a large cross-section, so it dominates the visible and near-IR. Its abundance $\propto n_e$, and in the photosphere the electrons come from metals; fewer metals means fewer electrons, less $\mathrm{H^-}$, lower opacity, so we see deeper layers.

Q3. Write the LTE bound–free absorption coefficient of hydrogen, including stimulated emission. Which quantities depend on temperature?

> **A3.** $\alpha_\nu^{\rm bf} = \sum_n N_n \sigma_{\rm bf}(n,\nu)\left(1 - e^{-h\nu/kT}\right)$, with $N_n = N_{\rm H\,I}\, 2n^2 e^{-E_n/kT}/U(T)$ and $N_{\rm H\,I}$ from the Saha equation. Temperature enters through the Boltzmann factor of each level, the Saha ionisation balance (through $T^{3/2}e^{-\chi/kT}/n_e$) and the stimulated-emission factor. The cross-sections themselves do not depend on T.

Q4. Rank the dominant continuous opacity in an O star, an A0 star, the Sun and an M dwarf, and say which produces a Balmer jump.

> **A4.** O star: electron scattering, with He bound–free. A0: H I bound–free and free–free. Sun: H^- bound–free (and free–free in the IR). M dwarf: H^- plus molecules. A strong Balmer jump appears where H I bound–free dominates, i.e. in late B, A and early F stars. It is weak in the Sun (H^- dominates) and in O stars (hydrogen ionised, electron scattering dominates).

Lecture SA3 — Physical Processes II: Transition Rates and Statistical Equilibrium

> **Sources:** D. Mihalas, *Stellar Atmospheres* (1978), the equations of statistical equilibrium and transition rates; Hubeny & Mihalas, *Theory of Stellar Atmospheres* (2014); Osterbrock & Ferland, *Astrophysics of Gaseous Nebulae* (2006) for collision strengths and critical densities; van Regemorter (1962), Seaton (1962) for collisional rate approximations. **Prerequisites:** RT4 (Einstein coefficients), SA1 (Saha–Boltzmann, two-level atom), SA2 (photo-ionisation cross-sections).

Learning goals

1. Write every radiative and collisional rate between bound levels and between a level and the continuum.
2. Derive the relations between forward and inverse rates (Einstein, Milne, Klein–Rosseland) from detailed balance.
3. Set up the statistical-equilibrium equations and recover LTE as a limit.
4. Use the **coronal** and **nebular** approximations, and compute critical densities from collision strengths.

Key concepts in plain words

A level population is a balance between traffic in and traffic out. Picture each atomic level as a room with doors to every other room and to the "continuum" (the ion plus a free electron). Every second, atoms leave a room by absorbing a photon, by emitting one, or by being hit by an electron. They enter it by the inverse processes. In a steady state, traffic in equals traffic out for every room: this is **statistical equilibrium**.

Two kinds of traffic. *Radiative* transitions depend on the radiation field (J_ν), which may come from far away and may not match the local temperature. *Collisional* transitions depend on the local electron density and temperature. Collisions push the populations toward LTE; radiation pushes them toward whatever the radiation field dictates.

Every process has an inverse, and they are linked. The probability of a process and of its inverse are related by atomic physics alone, independent of the environment. This is found by imagining the gas in full thermodynamic equilibrium, where each process must exactly balance its inverse (detailed balance). The Einstein relations (RT4) are one example; the same reasoning links photo-ionisation to recombination (Milne relation) and collisional excitation to collisional de-excitation (Klein–Rosseland relation).

Two simple limits. In very thin hot gas (the solar corona) collisions excite atoms but every excited atom decays by emitting a photon that escapes: the **coronal approximation**. In a nebula around a hot star, photo-ionisation by diluted starlight is balanced by recombination: the **nebular approximation**. Between these limits and LTE lies the full NLTE problem (SA7).

3.1 Inventory of rates

Number of transitions per unit volume per second from level i to level j: $n_i P_{ij}$, with $P_{ij} = R_{ij} + C_{ij}$.

Transition	Rate per atom	Inverse	Link (detailed balance)
radiative excitation $l \to u$	$R_{lu} = B_{lu}\bar{J}$	spontaneous + stimulated emission, $R_{ul} = A_{ul} + B_{ul}\bar{J}$	Einstein: $g_l B_{lu} = g_u B_{ul}$, $A_{ul} = \frac{2h\nu^3}{c^2} B_{ul}$
photo-ionisation $i \to \kappa$	$R_{i\kappa} = 4\pi \int_{\nu_i}^{\infty} \frac{\sigma_i(\nu)}{h\nu} J_\nu \, d\nu$	radiative recombination (spontaneous + stimulated)	Milne relation (below)
collisional excitation	$C_{lu} = n_e q_{lu}(T)$	collisional de-excitation $C_{ul} = n_e q_{ul}$	Klein–Rosseland: $\frac{q_{lu}}{q_{ul}} = \frac{g_u}{g_l} e^{-h\nu/kT}$
collisional ionisation	$C_{i\kappa} = n_e q_{i\kappa}(T)$	three-body recombination ($\propto n_e^2$)	Saha: $C_{\kappa i} = (n_i/n_\kappa)^* C_{i\kappa}$
—	—	dielectronic recombination	important in hot plasmas (corona, nebulae)
charge exchange	e.g. $O^+ + H \rightleftarrows O + H^+$	inverse reaction	detailed balance at T

Radiative recombination rate (the Milne relation in rate form). With $(n_i/n_\kappa)^*$ the LTE (Saha) ratio at the actual n_e and T:

$$n_\kappa R_{\kappa i} = n_\kappa \left(\frac{n_i}{n_\kappa}\right)^* 4\pi \int_{\nu_i}^{\infty} \frac{\sigma_i(\nu)}{h\nu} \left(\frac{2h\nu^3}{c^2} + J_\nu\right) e^{-h\nu/kT} \, d\nu .$$

If $J_\nu = B_\nu$ this exactly balances photo-ionisation, as detailed balance requires.

3.2 Collisional rates

Collisional rate coefficients are written with the dimensionless **collision strength** Ω_{lu} (Maxwell-averaged), symmetric in l and u:

$$q_{ul} = \frac{8.629 \times 10^{-6}}{g_u \sqrt{T}} \, \Omega_{lu} \ \mathrm{cm^3\,s^{-1}}, \qquad q_{lu} = \frac{g_u}{g_l} q_{ul} \, e^{-\Delta E/kT} = \frac{8.629 \times 10^{-6}}{g_l \sqrt{T}} \, \Omega_{lu} \, e^{-\Delta E/kT}.$$

(In SI multiply by 10^{-6}: 8.629×10^{-12} $\mathrm{m^3\,s^{-1}}$.) The relation between q_{lu} and q_{ul} is the **Klein–Rosseland relation**, the collisional analogue of the Einstein relations.

When no measured or computed Ω is available, approximations are used: **van Regemorter (1962)** for allowed transitions (relating Ω to the oscillator strength), **Seaton (1962)** for collisional ionisation, and **Drawin's formula** for collisions with neutral hydrogen in cool stars (often scaled by an empirical factor, now replaced by quantum calculations).

3.3 Statistical-equilibrium equations

For every level i (including the ground states of each ion):

$$\boxed{n_i \sum_{j \neq i} \left(R_{ij} + C_{ij}\right) = \sum_{j \neq i} n_j \left(R_{ji} + C_{ji}\right)}$$

Together with **particle conservation** ($\sum_i n_i = N_{\text{element}}$) and **charge conservation** ($n_e = \sum$ ion charges), this is a linear system for the populations *if* J_ν is known. It is written as a matrix equation $\mathsf{A}\,\mathbf{n} = \mathbf{b}$. The coupling with the RTE through J_ν makes the full problem non-linear (SA7).

LTE as a limit. If $C_{ij} \gg R_{ij}$ for all transitions, the equations reduce to collisional detailed balance, whose solution is Saha–Boltzmann at the local T. If $J_\nu = B_\nu(T)$, the radiative rates also balance pairwise. In both cases the populations are the LTE ones.

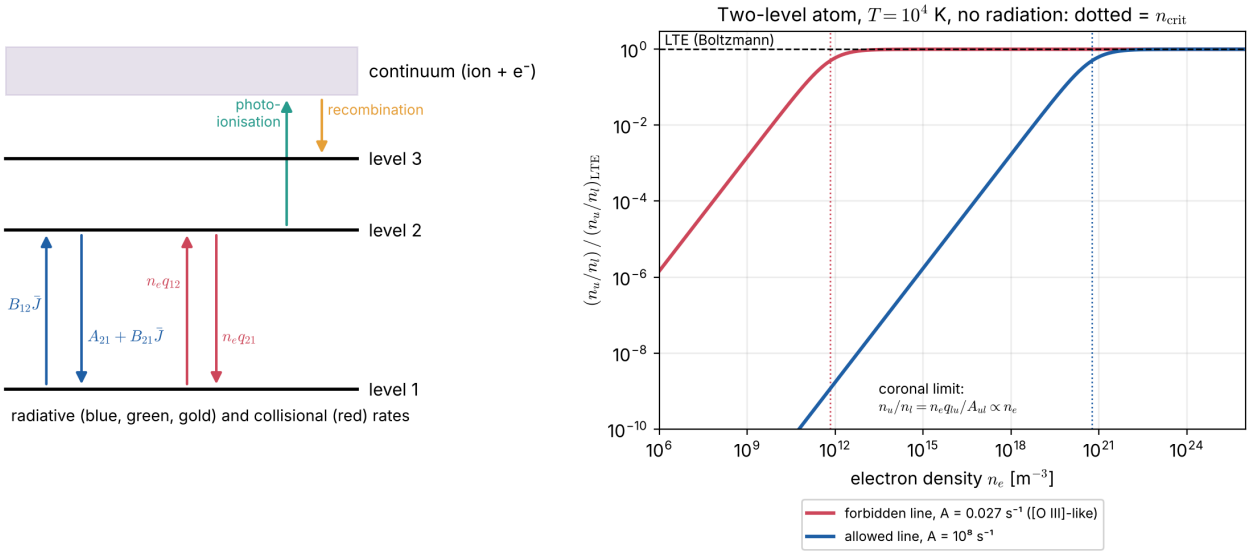


Figure 3.1 — Left: the radiative and collisional processes connecting bound levels and the continuum. Right: for an optically thin two-level atom with no radiation field ($T = 10^4$ K, $\Delta E = 2.48$ eV), the upper-level population relative to its LTE value rises $\propto n_e$ (coronal regime) and saturates at the Boltzmann value above the critical density (dotted). A forbidden line ($A = 0.027\ s^{-1}$) is in LTE above $\sim 7 \times 10^{11}\ m^{-3}$; an allowed line ($A = 10^8\ s^{-1}$) needs $\sim 6 \times 10^{20}\ m^{-3}$.

3.4 Two classical limits

Coronal approximation (hot, very low density, optically thin, weak radiation field; e.g. the solar corona, supernova remnants):

- excitation: $n_l n_e q_{lu} = n_u A_{ul}$, so $\dfrac{n_u}{n_l} = \dfrac{n_e q_{lu}}{A_{ul}} \ll$ Boltzmann, and the line emissivity $\propto n_e n_l q_{lu}$ (every collision gives one photon);
- ionisation: collisional ionisation = radiative + dielectronic recombination, so $\dfrac{n_{i+1}}{n_i} = \dfrac{q_{\rm ion}(T)}{\alpha_{\rm rec}(T)}$ is **independent of density** and depends on T only. This is why Fe IX–Fe XVI lines are coronal thermometers.

Nebular approximation (photo-ionised gas around a hot star, e.g. H II regions and planetary nebulae):

- the stellar radiation is diluted, $J_\nu \approx W\, B_\nu(T_*)$, with the dilution factor $W \approx \frac{1}{4}(R_*/r)^2 \ll 1$;

- ionisation balance: photo-ionisation from the ground state = recombination to all levels;
- excited levels are populated mainly by recombination cascades and collisions, and forbidden lines are collisionally excited with $n_e \ll n_{\rm crit}$ (SA7).

Worked example 3.1 — Collisional de-excitation from detailed balance

Starting from $n_l C_{lu} = n_u C_{ul}$ in TE, derive the Klein–Rosseland relation.

In TE the populations are Boltzmann, $n_u/n_l = (g_u/g_l)e^{-\Delta E/kT}$, and the electrons are Maxwellian at the same T. Collisional excitation must be exactly balanced by collisional de-excitation: $n_l n_e q_{lu} = n_u n_e q_{ul}$. So $q_{lu}/q_{ul} = n_u/n_l = (g_u/g_l)e^{-\Delta E/kT}$. Because q depends only on atomic physics and on the Maxwellian at T, the relation holds in any gas with Maxwellian electrons, even far from TE.

Worked example 3.2 — Critical density of [O III] 500.7 nm

The $^1D_2$ level of O^{2+} ($g_u = 5$) has total radiative decay rate $A \approx 0.027\ s^{-1}$, and the collision strength to the ground term is $\Omega \approx 2.29$. Compute q_{ul} and $n_{\rm crit}$ at $T = 10^4$ K.

$q_{ul} = 8.629 \times 10^{-12} \times 2.29/(5 \times 100) = 3.95 \times 10^{-14}$ m^3 s^{-1}; $n_{\rm crit} = A/q_{ul} = 6.8 \times 10^{11}$ m^{-3} (6.8×10^5 cm^{-3}). Figure 3.1 confirms that the upper level reaches 59 % of its LTE population at $n_e = 10^{12}$ m^{-3} but only 1.4 % at 10^{10} m^{-3} (typical of planetary nebulae).

Worked example 3.3 — An allowed line in the coronal regime

An allowed line with $A = 10^8\ s^{-1}$, $\Omega = 10$, $g_l = g_u = 5$, $\Delta E = 2.48$ eV at $T = 10^4$ K. Compute $n_{\rm crit}$ and $(n_u/n_l)/(n_u/n_l)_{\rm LTE}$ at $n_e = 10^{15}\ m^{-3}$ (solar transition region) and $10^{20}\ m^{-3}$ (deep photosphere).

$q_{ul} = 8.629 \times 10^{-12} \times 10/500 = 1.73 \times 10^{-13}$ m^3 s^{-1}, so $n_{\rm crit} = 10^8/1.73 \times 10^{-13} = 5.8 \times 10^{20}$ m^{-3}. Below $n_{\rm crit}$ the ratio is $\approx n_e/n_{\rm crit}$: 1.7×10^{-6} at 10^{15} m^{-3} (extreme NLTE, coronal) and 0.15 at 10^{20} m^{-3}. Even in a photosphere, strong allowed lines are not collisionally thermalised without help from photon trapping, which reduces the effective A by the escape probability (SA7).

Common misconceptions

1. **“Detailed-balance relations only apply in TE.”** They are derived in TE but involve only atomic properties (and the Maxwellian electrons), so they hold in any environment.
2. **“High temperature means LTE.”** The solar corona is very hot but far from LTE, because its density is tiny; what matters is the ratio of collisional to radiative rates.
3. **“Recombination is just the reverse arrow of photo-ionisation at the same rate.”** Only in LTE; in a nebula photo-ionisation is driven by diluted starlight while recombination proceeds at the local T_e, and the two balance at very different populations from Saha.
4. **“Coronal ionisation depends on density.”** In the coronal limit both ionisation and recombination are $\propto n_e$, so the ionisation balance depends on T alone.

Lecture summary

- Rates: radiative (bound–bound, bound–free) depend on J_ν; collisional depend on n_e and T.
- Detailed balance links each process to its inverse: Einstein (bound–bound radiative), Milne (bound–free radiative), Klein–Rosseland (collisional), Saha (collisional ionisation and three-body recombination).
- Statistical equilibrium: $n_i \sum_j P_{ij} = \sum_j n_j P_{ji}$ with particle and charge conservation; LTE when collisions dominate or $J = B$.
- Coronal limit: $n_u/n_l = n_e q_{lu}/A_{ul}$ and T-only ionisation balance. Nebular limit: diluted photo-ionisation balanced by recombination.
- Critical density $n_{\rm crit} = A/q_{ul}$, with $q_{ul} = 8.629 \times 10^{-6}\,\Omega/(g_u\sqrt{T})$ cm^3 s^{-1}.

Exam questions and answers — Lecture SA3

Q1. List the radiative and collisional processes that populate and depopulate a bound level, and give the detailed-balance relation for each pair.

> **A1.** Radiative excitation $B_{lu}\bar{J}$ ↔ spontaneous + stimulated emission $A_{ul} + B_{ul}\bar{J}$ (Einstein: $g_l B_{lu} = g_u B_{ul}$, $A_{ul} = 2h\nu^3 B_{ul}/c^2$). Photo-ionisation ↔ radiative recombination (Milne relation). Collisional excitation $n_e q_{lu}$ ↔ de-excitation $n_e q_{ul}$ (Klein–Rosseland: $q_{lu}/q_{ul} = (g_u/g_l)e^{-\Delta E/kT}$). Collisional ionisation ↔ three-body recombination (Saha ratio). Plus dielectronic recombination and charge exchange.

Q2. Derive the Klein–Rosseland relation and explain why it holds outside TE.

> **A2.** In TE, detailed balance gives $n_l q_{lu} = n_u q_{ul}$ with Boltzmann populations, so $q_{lu}/q_{ul} = (g_u/g_l)e^{-\Delta E/kT}$. The rate coefficients depend only on the cross-section and on the electron velocity distribution. As long as the electrons are Maxwellian at T, the relation holds whatever the radiation field or the level populations.

Q3. Describe the coronal approximation. Show that the ionisation balance is independent of density, and give one application.

> **A3.** Low density, optically thin gas with a weak radiation field: collisional excitations are balanced by spontaneous radiative decays ($n_l n_e q_{lu} = n_u A_{ul}$), and collisional ionisation by radiative and dielectronic recombination. Both ionisation and recombination rates per ion are $\propto n_e$: $n_i n_e q_{\rm ion} = n_{i+1} n_e \alpha_{\rm rec}$, so $n_{i+1}/n_i = q_{\rm ion}(T)/\alpha_{\rm rec}(T)$, independent of n_e. Application: Fe ion ratios (e.g. Fe IX–Fe XVI) measure coronal temperatures of 1–3 MK.

Q4. A forbidden line has $A = 0.02$ s^{-1}, $g_u = 5$ and $\Omega = 2$ at $T = 10^4$ K. Compute its critical density and state whether it is in LTE in a nebula with $n_e = 10^9$ m^{-3}.

> **A4.** $q_{ul} = 8.629 \times 10^{-12} \times 2/(5 \times 100) = 3.45 \times 10^{-14}$ m^3 s^{-1}; $n_{\rm crit} = 0.02/3.45 \times 10^{-14} = 5.8 \times 10^{11}$ m^{-3}. With $n_e = 10^9$ m^{-3} $\approx 2 \times 10^{-3} n_{\rm crit}$, the line is far from LTE: every collisional

> excitation produces a photon, and the upper level is underpopulated by a factor of about 600 relative to Boltzmann.

Lecture SA4 — Physical Processes III: Line Absorption and Line Broadening

> **Sources:** D. Mihalas, *Stellar Atmospheres* (1978), line absorption and spectral line broadening; D. F. Gray, *The Observation and Analysis of Stellar Photospheres* (2005), Chs. 11 (line absorption coefficient) and 18 (rotation); Rybicki & Lightman Ch. 10; Unsöld (1955) for van der Waals broadening; Anstee, Barklem & O'Mara (1995–2000, "ABO theory") for modern hydrogen-collision widths. **Prerequisites:** RT4 (Einstein coefficients), SA1 (populations), SA6 uses this lecture's profiles.

Learning goals

1. Write the line absorption coefficient in terms of the oscillator strength and a normalised profile.
2. Derive the natural (radiation-damping) width, and compute Doppler widths with microturbulence.
3. Classify pressure broadening by the power law of the interaction (C_n/r^n: linear Stark, resonance, quadratic Stark, van der Waals) and name the impact and quasi-static limits.
4. Build the Voigt profile, compute the damping parameter a, and add rotational and macroturbulent broadening.

Key concepts in plain words

Why a line has a width at all. An atomic transition has a definite energy, so in principle it absorbs at a single frequency. Three things smear it out. (1) *Uncertainty:* excited states live only a finite time, so their energy is slightly uncertain (natural broadening). (2) *Motion:* atoms move, so each one sees the light Doppler-shifted by a different amount (thermal and turbulent broadening). (3) *Neighbours:* nearby charged or neutral particles perturb the energy levels during collisions (pressure broadening). The first and third give a Lorentzian shape with broad wings; the second gives a Gaussian core. Their combination is the Voigt profile.

Two parts of a line tell different stories. The core of a strong line is shaped mainly by the Doppler effect, so it measures temperature and turbulence. The far wings are shaped by collisions, so they measure pressure, and through hydrostatic equilibrium the surface gravity. This is why the wings of Balmer lines and of the Mg I b and Ca II lines are used to measure $\log g$.

The star itself can broaden lines. Rotation makes one limb approach us and the other recede. Every line is then smeared by $\pm v \sin i$, with a characteristic rounded profile. Large-scale motions (macroturbulence, granulation) add further broadening. The instrument adds its own profile. All these broadenings act on the emergent spectrum, unlike the atomic ones, which act on the absorption coefficient.

4.1 The line absorption coefficient

For a transition $l \to u$ with oscillator strength f_{lu} and normalised profile $\phi(\nu)$ ($\int \phi\, d\nu = 1$), in LTE:

$$\alpha_\nu^{\rm line} = \frac{\pi e^2}{m_e c}\, f_{lu}\, n_l\, \phi(\nu)\left(1 - e^{-h\nu/kT}\right) \quad \text{(Gaussian units)}, \qquad \frac{\pi e^2}{m_e c} = 0.02654\ \text{cm}^2\,\text{Hz}.$$

The oscillator strength links to the Einstein coefficients: $g_l f_{lu} = \dfrac{m_e c^3}{8\pi^2 e^2 \nu^2}\, g_u A_{ul}$, so $A_{ul} = 0.6670\, \dfrac{g_l}{g_u}\, \dfrac{f_{lu}}{\lambda^2}\ \text{s}^{-1}$ with λ in cm. Spectroscopists usually quote $\log gf \equiv \log(g_l f_{lu})$.

4.2 Natural (radiation) broadening

An excited state decaying with total rate $\Gamma_u = \sum_{j<u} A_{uj}$ has a Lorentzian energy distribution. For a transition between two levels, both of which decay:

$$\phi(\nu) = \frac{\gamma/4\pi^2}{(\nu - \nu_0)^2 + (\gamma/4\pi)^2}, \qquad \gamma_{\rm rad} = \Gamma_u + \Gamma_l, \qquad \Delta\lambda_{\rm FWHM} = \frac{\lambda^2 \gamma}{2\pi c}.$$

The classical (damped-oscillator) value $\gamma_{\rm cl} = 8\pi^2 e^2 \nu^2/(3 m_e c^3) = 2.47 \times 10^{-22} \nu^2\ \text{s}^{-1}$ is a useful order-of-magnitude estimate.

4.3 Doppler broadening

With a Maxwellian distribution plus Gaussian microturbulence ξ:

$$\phi_D(\nu) = \frac{1}{\sqrt{\pi}\,\Delta\nu_D} e^{-(\nu-\nu_0)^2/\Delta\nu_D^2}, \qquad \Delta\nu_D = \frac{\nu_0}{c}\sqrt{\frac{2kT}{m} + \xi^2}.$$

Light atoms (H) have broad Doppler cores; heavy atoms (Fe) narrow ones, so microturbulence matters most for metal lines.

4.4 Pressure (collisional) broadening

A perturber at distance r shifts the line frequency by $\Delta\nu = C_n / r^n$:

n	Name	Perturbers	Where it matters
2	**linear Stark**	ions and electrons	hydrogen and He II lines in A, B stars and white dwarfs (very broad Balmer wings)
3	**resonance** (self) broadening	atoms of the same species	lines connected to the ground state, e.g. Balmer lines in cool stars
4	**quadratic Stark**	electrons, ions	non-hydrogenic atoms in hot stars
6	**van der Waals**	neutral H (and He, H_2)	metal lines in cool stars (Sun: Na D, Mg I b wings)

Two limits of the theory:

- **impact approximation:** collisions are brief and separated. They interrupt the radiation, giving a **Lorentzian** with width $\gamma_n \propto N_{\rm perp}\, C_n^{2/(n-1)}\, \bar{v}^{(n-3)/(n-1)}$ (for van der Waals, $\gamma_6 \propto N_{\rm H} C_6^{2/5} \bar{v}^{3/5}$, Unsöld's classical estimate);
- **quasi-static approximation:** perturbers barely move during the emission. The statistical distribution of their fields gives the profile (the **Holtsmark** distribution for linear Stark broadening by ions).

Modern van der Waals widths use the **ABO theory** (Anstee, Barklem & O'Mara), which corrects Unsöld's values by factors of about 1.5–2.5.

4.5 The Voigt profile

Natural plus pressure broadening (Lorentzian, total $\gamma = \gamma_{\rm rad} + \gamma_4 + \gamma_6 + \ldots$) convolved with Doppler broadening (Gaussian):

$$\phi(\nu) = \frac{H(a,v)}{\sqrt{\pi}\,\Delta\nu_D}, \qquad H(a,v) = \frac{a}{\pi}\int_{-\infty}^{\infty} \frac{e^{-y^2}}{(v-y)^2 + a^2}\,dy, \qquad a = \frac{\gamma}{4\pi\Delta\nu_D}, \quad v = \frac{\nu-\nu_0}{\Delta\nu_D}.$$

Core: $H \approx e^{-v^2}$. Far wings: $H \approx a/(\sqrt{\pi} v^2)$. The wings take over at $|v| \approx 2$–3 for $a \sim 0.01$.

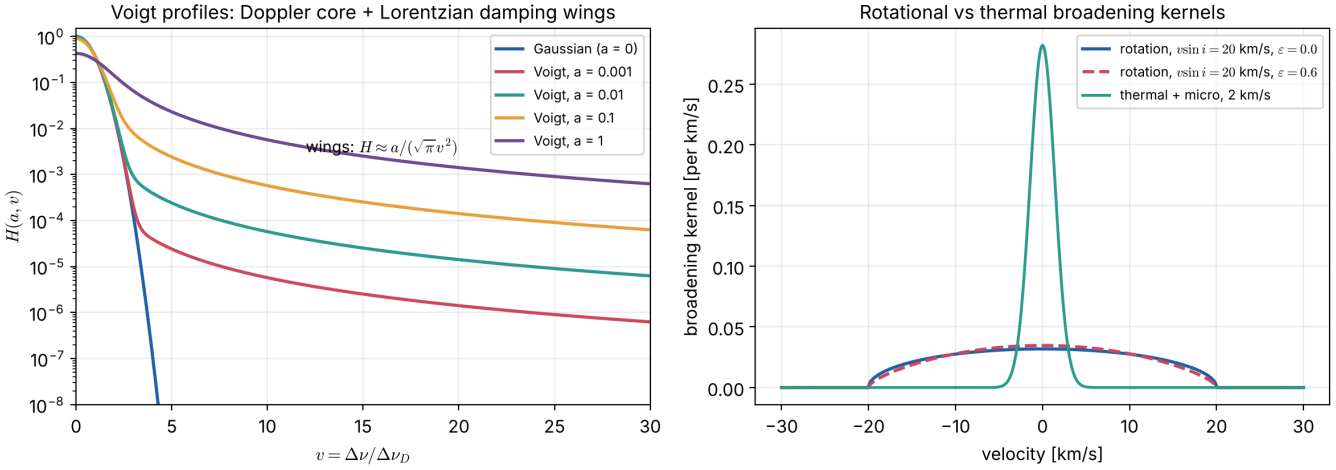


Figure 4.1 — Left: Voigt functions for increasing damping a; the Lorentzian wings rise above the Gaussian core beyond $v \approx 2$–3. Right: the rotational broadening kernel for $v \sin i = 20$ km s^{-1}, without and with limb darkening ($\varepsilon = 0.6$), compared with a 2 km s^{-1} thermal-plus-microturbulent Gaussian.

4.6 Broadening of the emergent spectrum

- **Rotation:** the flux profile is convolved with the kernel $G(\Delta\lambda) = \dfrac{2(1-\varepsilon)\sqrt{1-x^2} + \frac{\pi\varepsilon}{2}(1-x^2)}{\pi\Delta\lambda_L(1-\varepsilon/3)}$, with $x = \Delta\lambda/\Delta\lambda_L$, $\Delta\lambda_L = \lambda v \sin i/c$, and ε the linear limb-darkening coefficient (Gray, Ch. 18).
- **Macroturbulence:** large-scale motions (granulation), modelled with a radial–tangential profile; 3D models produce it naturally.
- **Instrumental profile:** set by the resolving power $R = \lambda/\Delta\lambda$.
- **Zeeman splitting** (RT9) and hyperfine or isotopic structure.

Worked example 4.1 — Natural width of Hα

For Hα (656.28 nm) the upper level $n = 3$ decays with $A_{31} + A_{32} \approx 5.58 \times 10^7 + 4.41 \times 10^7$ s^{-1}, and the lower level $n = 2$ decays by Lyα with $A_{21} = 6.27 \times 10^8$ s^{-1}. Compute the natural FWHM and compare with the Doppler width at 5800 K (0.21 Å, SA6).

$\gamma = 1.0 \times 10^8 + 6.27 \times 10^8 = 7.3 \times 10^8$ s^{-1}, so $\Delta\lambda = \lambda^2\gamma/(2\pi c) = (6.5628 \times 10^{-7})^2 \times 7.3 \times 10^8/(2\pi \times 3 \times 10^8) = 1.7 \times 10^{-13}$ m $= 0.0017$ Å. This is about 120 times narrower than the Doppler core. Natural broadening matters only in the far wings, and only where pressure broadening is weak.

Worked example 4.2 — Damping parameter of a solar iron line

An Fe I line at 500 nm in the Sun has a total damping constant (mostly van der Waals) $\gamma \approx 10^9$ s^{-1}. With $T = 5772$ K and $\xi = 1$ km s^{-1}, compute a.

$v_D = \sqrt{2kT/m_{\mathrm{Fe}} + \xi^2} = 1.65$ km s^{-1}, $\Delta\nu_D = v_D/\lambda = 3.3 \times 10^9$ Hz, so $a = \gamma/(4\pi\Delta\nu_D) = 10^9/(4\pi \times 3.3 \times 10^9) = 0.024$. This is why $a \approx 0.01$–0.03 is used for solar metal lines (and in the curve of growth of SA6).

Worked example 4.3 — When does rotation dominate?

Compare the rotational broadening of a 500 nm line for $v \sin i = 2$ km s^{-1} (the Sun) and 50 km s^{-1} (a typical A star) with the Fe Doppler width.

$\Delta\lambda_L = \lambda v \sin i/c$: 2 km s^{-1} → 0.0033 nm (33 mÅ), comparable to the Doppler width (about 27 mÅ including microturbulence); 50 km s^{-1} → 0.083 nm (0.83 Å), 30 times larger. In rapid rotators the line shapes are dominated by rotation, weak lines blend together, and abundance analysis becomes harder.

Common misconceptions

1. **“The natural width is what we see.”** It is usually 10–1000 times smaller than the Doppler and pressure widths; it matters only in the far wings.
2. **“All broadening affects the absorption coefficient.”** Rotation, macroturbulence and the instrument broaden the *emergent* spectrum but leave the equivalent width unchanged; atomic broadening changes the absorption profile and hence, for saturated lines, the equivalent width.
3. **“Pressure broadening is the same for all lines.”** It depends on the type of interaction (Stark, van der Waals) and on the perturbers (electrons, ions, neutral H), so different lines respond differently to gravity.
4. **“The Voigt core is Lorentzian.”** The core is essentially Gaussian; the Lorentzian shows only in the wings.

Lecture summary

- $\alpha_\nu^{\mathrm{line}} = (\pi e^2/m_e c) f_{lu} n_l \phi(\nu)(1 - e^{-h\nu/kT})$; $\log gf$ measures line strength.
- Natural: Lorentzian with $\gamma_{\mathrm{rad}} = \Gamma_u + \Gamma_l$. Doppler: Gaussian with $\Delta\nu_D = (\nu_0/c)\sqrt{2kT/m + \xi^2}$.
- Pressure: C_n/r^n with $n = 2, 3, 4, 6$ (linear Stark, resonance, quadratic Stark, van der Waals); impact (Lorentz) versus quasi-static (Holtsmark) limits; ABO theory for van der Waals.
- Voigt: $H(a, v)$ with $a = \gamma/(4\pi\Delta\nu_D)$; Gaussian core, $a/(\sqrt{\pi}v^2)$ wings.
- Emergent-spectrum broadening: rotation ($v \sin i$), macroturbulence, instrument, Zeeman.

Exam questions and answers — Lecture SA4

Q1. Name the three atomic broadening mechanisms and the profile shape each produces. Which part of a strong line does each control?

> **A1.** Natural (finite lifetimes) gives a Lorentzian; pressure/collisional (perturbing particles) gives a Lorentzian in the impact limit, or a Holtsmark-type profile in the quasi-static limit; Doppler (thermal and turbulent motion) gives a Gaussian. The convolution is the Voigt profile. The Doppler part controls the core (temperature, microturbulence); the Lorentzian parts control the far wings (pressure, hence gravity, and atomic data).

Q2. Define the damping parameter a and compute it for a line at 600 nm with $\gamma = 5 \times 10^8$ s^{-1}, $T = 6000$ K, atomic mass 40 and $\xi = 0$.

> **A2.** $a = \gamma / (4\pi \Delta\nu_D)$. $v_D = \sqrt{2kT/m} = \sqrt{2 \times 1.381 \times 10^{-23} \times 6000 / (40 \times 1.661 \times 10^{-27})} = 1.58$ km s^{-1}; $\Delta\nu_D = v_D / \lambda = 2.63 \times 10^9$ Hz; $a = 5 \times 10^8 / (4\pi \times 2.63 \times 10^9) = 0.015$.

Q3. Which pressure-broadening mechanism dominates (a) the Balmer lines of an A0 star, (b) the Na D wings in the Sun, (c) Si III lines in a B star? Explain.

> **A3.** (a) Linear Stark broadening by ions and electrons: hydrogen has degenerate levels, so the first-order Stark effect acts, and A stars have abundant free charges. (b) Van der Waals broadening by neutral hydrogen: the solar photosphere is mostly neutral H, and Na is non-hydrogenic. (c) Quadratic Stark broadening by electrons: a non-hydrogenic ion in a hot, ionised atmosphere.

Q4. Why do the wings of strong lines measure surface gravity? Give the chain of reasoning.

> **A4.** Wing strength $\propto \gamma_{\rm pressure} \propto$ the number density of perturbers (electrons, ions or neutral H) where the wings form. By hydrostatic equilibrium the pressure at a given optical depth scales as $P \approx g\tau/\kappa$, so higher g means higher density, larger damping, and broader wings at fixed $T_{\rm eff}$ and abundance. Comparing observed wing profiles (Balmer lines in hot stars, Mg I b or Ca II in cool stars) with synthetic spectra therefore gives $\log g$.

Lecture SA5 — The Gray LTE Atmosphere: Complete Solution

This is the central lecture of the course. We solve, step by step and then exactly, the simplest self-consistent model of a stellar atmosphere in LTE. **Sources:** Collins §9.1d (assumptions), §9.4 (radiative equilibrium; Eddington approximation, eq. 9.4.12–9.4.13); Chandrasekhar Ch. III §25 (Milne problem by discrete ordinates: $J(\tau) = \frac{3}{4}F[\tau + q(\tau)]$, eq. 45; H-function law of darkening, eq. 61; Hopf–Bronstein relation $J(0) = \frac{\sqrt{3}}{4}F$, eq. 64) and Ch. V (exact tables); Hu Set 1; MIT 8.901.

Learning goals

1. List the assumptions of the gray LTE atmosphere and explain each.
2. Derive $T^4(\tau) = \frac{3}{4}T_{\text{eff}}^4\left(\tau + \frac{2}{3}\right)$ in the Eddington approximation.
3. Know the exact solution (Hopf function $q(\tau)$, Chandrasekhar's H-function) and its numerical values.
4. Compute the emergent intensity, limb darkening, spectrum and pressure structure of a gray LTE model.

Key concepts in plain words

The "hydrogen atom" of stellar atmospheres. The gray atmosphere is the simplest model that is still physically self-consistent: one opacity for all frequencies, LTE, a flat geometry, and all energy carried by radiation. Like the hydrogen atom in quantum mechanics, it can be solved exactly, and its solution explains the main features of real stars: why they are hotter inside, what T_{eff} means, why their disks are limb-darkened, and why their spectra are close to (but not exactly) black bodies.

Why $J = B$ in a gray LTE atmosphere. Radiative equilibrium says each layer emits as much as it absorbs. With the same opacity at all frequencies, "absorbed = emitted" becomes simply "the average radiation bathing the gas (J) equals what the gas emits per unit absorption (B)". So the gas temperature is set by how strongly it is illuminated: $\sigma T^4/\pi = J$.

Why the temperature rises inward. The flux must be the same at every depth (no energy is created or lost). Carrying a constant flux outward through an opaque medium requires a gradient, exactly as heat flows from hot to cold only if there is a temperature difference. The deeper you go, the more optical depth the energy must still cross, so the hotter the gas must be: $T^4 \propto \tau +$ constant.

Why the surface is not at zero temperature. At the top, the gas still receives radiation from below (about half the sky is "bright"), so it stays warm: $T(0) \approx 0.81$–$0.84\,T_{\text{eff}}$. It cannot cool to zero, because it is constantly heated by the light passing through it.

What T_{eff} really means. The effective temperature is defined from the total flux, $F = \sigma T_{\text{eff}}^4$. The gray solution shows that it equals the actual gas temperature at $\tau \approx 2/3$. This is exactly the depth from which, by Eddington–Barbier, the flux escapes. "The photosphere" is therefore the layer at $\tau \approx 2/3$, where the gas temperature equals T_{eff}.

Approximate versus exact. Eddington's approximation gives $T^4 = \frac{3}{4}T_{\rm eff}^4(\tau + \frac{2}{3})$. The exact solution replaces the constant $\frac{2}{3}$ by the slowly varying Hopf function $q(\tau)$, between 0.577 at the top and 0.710 deep down. The difference is only a few per cent in temperature, and it matters mainly near the surface. Our numerical solver reproduces the exact values of Chandrasekhar's 1950 tables.

Why the spectrum is not a black body. Each frequency sees a mixture of layers at different temperatures. Because the Planck function rises very steeply with T in the ultraviolet, the hot deeper layers contribute disproportionately there, so a gray star is slightly brighter than a black body in the UV. The **total** flux, though, is exactly $\sigma T_{\rm eff}^4$.

5.1 The problem

Assumptions

1. **Plane-parallel, static** atmosphere; τ measured inward.
2. **LTE:** $S_\nu = B_\nu(T)$ (pure absorption; scattering can be added and changes nothing for the gray case, since it cancels in radiative equilibrium).
3. **Gray:** $\kappa_\nu = \kappa$ is independent of frequency, so $\tau_\nu = \tau$ for all ν. (In practice we use the Rosseland mean: Lecture RT8.)
4. **Radiative equilibrium:** constant flux $F = \sigma T_{\rm eff}^4$, with no convection and no energy sources.

Frequency-integrated equations. Integrating $\mu\, dI_\nu/d\tau = I_\nu - B_\nu$ over ν (allowed because τ does not depend on ν):

$$\mu \frac{dI}{d\tau} = I - B, \qquad B = \frac{\sigma T^4}{\pi}.$$

5.2 Step-by-step solution in the Eddington approximation

Step 1 — Radiative equilibrium gives $J = B$. Radiative equilibrium, $\int \kappa_\nu (J_\nu - B_\nu) d\nu = 0$, becomes for a gray atmosphere $\kappa(J - B) = 0$:

$$\boxed{J(\tau) = B(\tau) = S(\tau)}$$

In a gray LTE atmosphere the source function equals the mean intensity. This is mathematically identical to a **pure-scattering** atmosphere (the Milne problem), which is why Chandrasekhar treats both together.

Step 2 — The flux is constant. The zeroth moment gives $dH/d\tau = J - B = 0$, so

$$H = \text{const} = \frac{F}{4\pi} = \frac{\sigma T_{\rm eff}^4}{4\pi}.$$

Step 3 — K is linear in τ (exactly). The first moment, $dK/d\tau = H$, gives

$$K(\tau) = H\tau + \text{const}.$$

No approximation has been made so far.

Step 4 — Eddington closure. With $K = J/3$:

$$J(\tau) = 3H\,(\tau + C).$$

Step 5 — Surface boundary condition. With $J(0) = 2H(0)$ (Lecture RT8): $3HC = 2H$, so $C = 2/3$:

$$J(\tau) = 3H\left(\tau + \frac{2}{3}\right) = \frac{3\sigma T_{\text{eff}}^4}{4\pi}\left(\tau + \frac{2}{3}\right).$$

Step 6 — Temperature. Since $J = B = \sigma T^4/\pi$:

$$\boxed{T^4(\tau) = \frac{3}{4}\,T_{\text{eff}}^4\left(\tau + \frac{2}{3}\right)}$$

Step 7 — Consequences.

Quantity	Value	Meaning
$T(\tau = 2/3)$	T_{eff}	the "photosphere": the effective temperature is the actual temperature at $\tau = 2/3$ (Eddington–Barbier)
$T(0)$	$2^{-1/4}T_{\text{eff}} = 0.841\,T_{\text{eff}}$	**boundary temperature**; the atmosphere does not cool to zero
$\tau \gg 1$	$T^4 \approx \frac{3}{4}T_{\text{eff}}^4\tau$	diffusion regime, $F = \frac{4\pi}{3}\frac{dB}{d\tau}$ ✓

5.3 Emergent intensity and limb darkening

Since $S(\tau) = \frac{3F}{4\pi}\left(\tau + \frac{2}{3}\right)$ is **linear**, Eddington–Barbier is exact:

$$I(0,\mu) = S(\tau = \mu) = \frac{3F}{4\pi}\left(\mu + \frac{2}{3}\right) \qquad \Rightarrow \qquad \boxed{\frac{I(0,\mu)}{I(0,1)} = \frac{2 + 3\mu}{5} = \frac{2}{5}\left(1 + \frac{3}{2}\mu\right)}$$

Self-consistency check. The flux computed from the emergent intensity is

$$2\pi\int_0^1 I(0,\mu)\,\mu\,d\mu = 2\pi\cdot\frac{3F}{4\pi}\left(\frac{1}{3} + \frac{1}{3}\right) = F. \checkmark$$

5.4 The exact solution (Milne, Hopf, Chandrasekhar)

Without the Eddington approximation, Step 1 combined with the Schwarzschild–Milne equation (Lecture RT7) gives **Milne's integral equation**

$$J(\tau) = \Lambda_\tau[J] = \frac{1}{2}\int_0^\infty J(t)\,E_1|t - \tau|\,dt.$$

Its solution, normalised to flux F, is written (Chandrasekhar eq. 45)

$$\boxed{J(\tau) = \frac{3F}{4\pi}\left[\tau + q(\tau)\right], \qquad T^4 = \frac{3}{4}T_{\text{eff}}^4\left[\tau + q(\tau)\right]}$$

where $q(\tau)$ is the **Hopf function**:

τ	0	0.01	0.1	0.5	1	2	∞
$q(\tau)$ (our solver)	0.5773	0.5882	0.6279	0.6802	0.6985	0.7078	0.7104
exact	$1/\sqrt{3} = 0.57735$						0.710446

- $q(0) = 1/\sqrt{3}$ is the **Hopf–Bronstein relation** $J(0) = \sqrt{3}\,H$ (Chandrasekhar eq. 64), an exact result.
- Exact boundary temperature: $T(0) = (\sqrt{3}/4)^{1/4} T_{\text{eff}} = 0.8114\,T_{\text{eff}}$, against Eddington's 0.8409 (a 3.6 % error).
- $q(\infty) = 0.7104$ against Eddington's $2/3$: the deep structure is shifted by $\Delta\tau = 0.044$.

Exact emergent intensity (Chandrasekhar eq. 61): $I(0,\mu) = \dfrac{\sqrt{3}}{4}\dfrac{F}{\pi} H(\mu)$, where $H(\mu)$ solves

$$\frac{1}{H(\mu)} = \int_0^1 \frac{\mu' H(\mu')}{2(\mu+\mu')}\,d\mu', \qquad H(0) = 1,\ H(1) = 2.9078.$$

μ	0	0.1	0.2	0.5	1
exact $I(\mu)/I(1)$ (Chandrasekhar Table; our solver)	0.3439	0.4290	0.4988	0.6922	1
Eddington $(2+3\mu)/5$	0.400	0.460	0.520	0.700	1

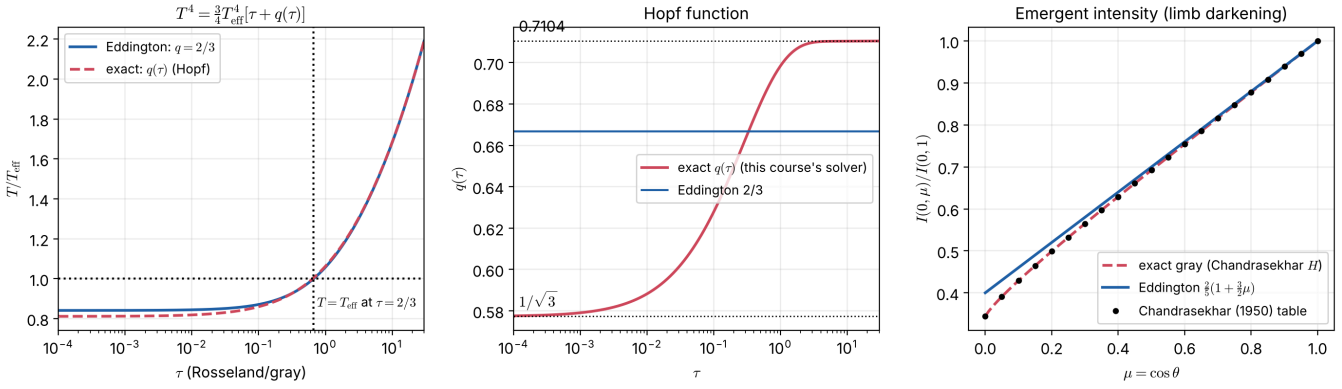


Figure 5.1 — Left: temperature structure, Eddington versus exact. They differ only for $\tau \lesssim 1$. Centre: the Hopf function from our numerical solution of Milne's equation. Right: bolometric limb darkening. The exact H-function law (dashed) agrees with Chandrasekhar's 1950 table (dots) to 4 decimals; the Eddington law (solid) is too bright at the limb.

5.5 Numerical solution of the gray LTE problem

`scripts/lte_solvers.py → hopf_function()` solves Milne's equation directly:

1. Grid τ_i from 0 to 30 (logarithmic).
2. Build the Λ matrix analytically (piecewise-linear S; Lecture RT8).
3. Below $\tau_{\max}$, attach the known asymptotic form $S = \frac{3}{4}(\tau + q_\infty)$ and treat its contribution as a known right-hand side.
4. Solve $(\mathbf{1} - \mathbf{\Lambda})\mathbf{S} = \mathbf{b}$; then $q = \frac{4}{3}S - \tau$.

Results: $q(0) = 0.5773$ (exact 0.57735), $q(1) = 0.6985$. The emergent flux, integrated over all wavelengths with $T(\tau)$ from this solution and the LTE formal solution $F_\lambda = 2\pi \int_0^\infty B_\lambda(T(\tau)) E_2(\tau) d\tau$, equals $\sigma T_{\rm eff}^4$ to **0.007 %**, confirming radiative equilibrium.

5.6 Monochromatic properties: the spectrum of a gray LTE star

Although κ is gray, the **spectrum is not a black body**, because each frequency samples $B_\nu(T(\tau))$ with T varying with depth:

$$I_\nu(0,\mu) = \int_0^\infty B_\nu\big(T(\tau)\big)\, e^{-\tau/\mu}\, \frac{d\tau}{\mu}, \qquad F_\nu(0) = 2\pi \int_0^\infty B_\nu\big(T(\tau)\big) E_2(\tau)\, d\tau.$$

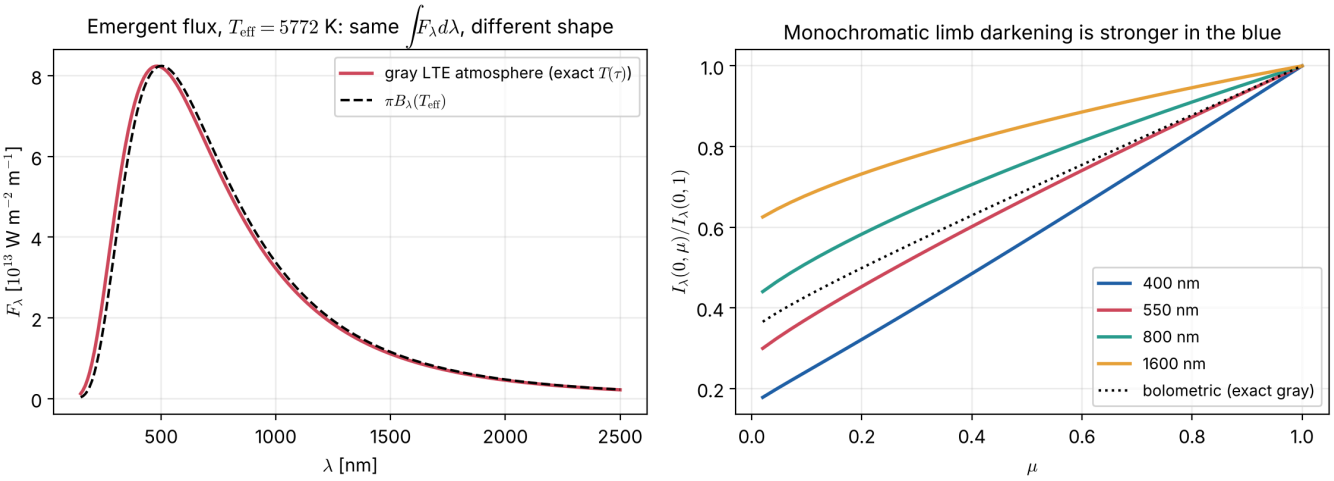


Figure 5.2 — Left: emergent flux of the exact gray LTE atmosphere with $T_{\rm eff} = 5772$ K compared with $\pi B_\lambda(T_{\rm eff})$. The UV excess comes from hot deep layers that shine through. Right: limb darkening is much stronger in the blue, because B_λ is more sensitive to T in the Wien regime (Lecture RT3).

λ	$I(0.1)/I(1)$	$I(0.5)/I(1)$
400 nm	0.242	0.569
550 nm	0.372	0.672
800 nm	0.510	0.761

5.7 Adding hydrostatic equilibrium: a complete model

With $dP/dz = -\rho g$ and $d\tau = -\kappa\rho\, dz$:

$$\frac{dP}{d\tau} = \frac{g}{\kappa} \quad \Rightarrow \quad P(\tau) = \frac{g}{\kappa}\tau \quad (\kappa \text{ constant}).$$

Then $\rho = P\mu m_H/(kT)$. Together with $T(\tau)$ this gives a full **gray LTE model atmosphere**: $T(\tau)$, $P(\tau)$, $\rho(\tau)$, as functions of only $T_{\rm eff}$, g and κ (i.e. composition), exactly as Collins' five assumptions promised (Lecture SA1).

Real (non-gray) atmospheres need iteration: guess $T(\tau)$ → Saha–Boltzmann gives κ_ν → solve the RTE at all ν → check $\int \kappa_\nu(J_\nu - B_\nu)d\nu = 0$ and flux constancy → correct T (Unsöld–Lucy correction) → repeat. The gray solution is the standard **starting model**.

Worked example 5.1 — The Sun as a gray LTE atmosphere

$T_{\rm eff} = 5772$ K. *Find* T *at* $\tau = 0, 2/3, 1, 2, 10$ *(Eddington), and the exact surface temperature.*

τ	0	2/3	1	2	10
T (Eddington) [K]	4854	5772	6103	6864	9707

Exact $T(0) = 0.8114 \times 5772 = 4682$ K. The observed solar temperature minimum is about 4400 K (it then rises in the chromosphere, which a radiative-equilibrium model cannot reproduce). Compare Lecture RT7: the limb-darkening inversion at 500 nm gave 5155 K at $\tau_{500} = 0$ and 6300 K at $\tau_{500} = 1$. The gray model is the right order of magnitude, with differences due to non-grayness (H^- opacity varies with λ).

Worked example 5.2 — Pressure at the solar photosphere

Take $g_\odot = 274$ *m s*$^{-2}$ *and a mean opacity* $\kappa \approx 0.03$ *m*^{2} *kg*$^{-1}$. *Estimate* P *and* ρ *at* $\tau = 2/3$.

$P = g\tau/\kappa = 274 \times 0.667/0.03 = 6.1 \times 10^3$ Pa (about 0.06 atm). With $\mu \approx 1.3$ (neutral gas) and $T = 5772$ K: $\rho = P\mu m_H/(kT) = 6.1 \times 10^3 \times 1.3 \times 1.67 \times 10^{-27}/(1.38 \times 10^{-23} \times 5772) = 1.7 \times 10^{-4}$ kg m^{-3}. Detailed solar models give $\rho \approx 2.5 \times 10^{-4}$ kg m^{-3} at $\tau_{500} = 1$, so the gray estimate is good to a factor of order unity.

Worked example 5.3 — Verifying Step 1 directly

Show that if $S = J$ *and* H = const, *then* $\int_0^\infty J(t)E_2(t)dt = 2H$ *(the flux from the Schwarzschild–Milne equation at* $\tau = 0$*), and check it for the Eddington solution.*

$F(0) = 2\pi \int_0^\infty S(t)E_2(t)dt = 4\pi H \Rightarrow \int_0^\infty SE_2\,dt = 2H$. Eddington: $S = 3H(t + 2/3)$, and $\int_0^\infty tE_2dt = 1/3$, $\int_0^\infty E_2dt = E_3(0) = 1/2$. So $\int SE_2 = 3H(1/3 + 1/3) = 2H$ ✓. The Eddington solution satisfies the **surface** flux condition exactly, but not Milne's equation at every depth. The exact $q(\tau)$ corrects that.

Common misconceptions

1. **"$T_{\rm eff}$ is the temperature of the stellar surface."** It is the temperature at $\tau \approx 2/3$; the top of the atmosphere is cooler (about $0.81\,T_{\rm eff}$).
2. **"A gray atmosphere emits a black-body spectrum at $T_{\rm eff}$."** Its spectrum is a blend of Planck functions from many depths.
3. **"The Eddington solution satisfies the transfer equation exactly."** It satisfies the moment equations with an approximate closure; the exact solution needs the Hopf function.
4. **"Gray means the opacity is small."** Gray means *independent of frequency*, whatever its size.

Lecture summary

- Gray + LTE + radiative equilibrium ⇒ $J = B$, H = constant, $K = H\tau +$ const.
- Eddington: $T^4 = \frac{3}{4}T_{\rm eff}^4(\tau + \frac{2}{3})$; $T(0) = 0.841\,T_{\rm eff}$; $T = T_{\rm eff}$ at $\tau = 2/3$.
- Exact: $q(\tau)$ from 0.577 to 0.710; $T(0) = 0.811\,T_{\rm eff}$; limb darkening $H(\mu)/H(1)$.

- Emergent spectrum is not a black body; total flux $= \sigma T_{\rm eff}^4$; add $P = g\tau/\kappa$ for a full model.

Exam questions and answers — Lecture SA5

Q1. State the assumptions of the gray LTE atmosphere and derive $T^4 = \frac{3}{4}T_{\rm eff}^4(\tau + \frac{2}{3})$. Indicate clearly where the Eddington approximation enters.

> **A1.** Assumptions: plane-parallel, static, LTE ($S = B$), gray ($\kappa_\nu = \kappa$), radiative equilibrium with $F = \sigma T_{\rm eff}^4$. Derivation: (i) RE with gray opacity gives $J = B$; (ii) $dH/d\tau = J - B = 0 \Rightarrow H = F/4\pi$; (iii) $dK/d\tau = H \Rightarrow K = H\tau + c$ (exact). **(iv) Eddington:** $K = J/3 \Rightarrow J = 3H(\tau + C)$. **(v) Eddington boundary condition** $J(0) = 2H$ gives $C = 2/3$. (vi) $J = B = \sigma T^4/\pi$ and $H = \sigma T_{\rm eff}^4/4\pi$ give $T^4 = \frac{3}{4}T_{\rm eff}^4(\tau + \frac{2}{3})$.

Q2. Derive the limb-darkening law $I(\mu)/I(1) = \frac{2}{5}(1 + \frac{3}{2}\mu)$ and verify that it gives the correct flux.

> **A2.** $S = 3H(\tau + 2/3)$ is linear, so $I(0,\mu) = S(\mu) = 3H(\mu + 2/3)$, and $I(\mu)/I(1) = (\mu + 2/3)/(5/3) = (2 + 3\mu)/5$. Flux: $2\pi \int_0^1 3H(\mu + \frac{2}{3})\mu\, d\mu = 6\pi H(\frac{1}{3} + \frac{1}{3}) = 4\pi H = F$ ✓.

Q3. What is the Hopf function? Give $q(0)$ and $q(\infty)$, and explain how the Eddington solution approximates it.

> **A3.** The exact solution of Milne's equation is $J = 3H[\tau + q(\tau)]$, where q increases monotonically from $q(0) = 1/\sqrt{3} = 0.577$ (Hopf–Bronstein) to $q(\infty) = 0.7104$. Eddington replaces it by the constant $2/3 = 0.667$, which lies between the two limits; the maximum error in T is 3.6 %, at the surface.

Q4. Explain why the emergent spectrum of a gray LTE atmosphere is not a black body at $T_{\rm eff}$, and why it shows an excess in the UV.

> **A4.** $F_\nu = 2\pi \int B_\nu(T(\tau))E_2(\tau)d\tau$ is a weighted average of Planck functions over a range of temperatures. Because B_ν is a convex, steep function of T in the Wien (UV) regime, the hotter layers contribute disproportionately there, so the UV flux exceeds $\pi B_\nu(T_{\rm eff})$. Energy conservation then requires a slight deficit elsewhere. The total $\int F_\nu d\nu$ equals $\sigma T_{\rm eff}^4$ exactly.

Lecture SA6 — Spectral Lines in LTE, and the Limits of LTE

> **Sources:** Collins §9.1c (continuum and spectral lines), §9.2e (redistribution: coherent, complete, isotropic); Chandrasekhar Ch. XII §83–84 (line formation; Milne–Eddington model, residual intensities); MIT 8.901 Lec. 7 §7.3 (two-layer atmosphere, thermal inversions); Hu Set 1 ("photons escape through the channel of lowest opacity, so lines are dark"); Lectures RT4, RT7 (Radiative Transfer course) and SA1, SA5 (this course).

Learning goals

1. Write the line absorption coefficient, the line profile (Doppler, Lorentz, Voigt) and the Doppler width.
2. Explain with Eddington–Barbier why LTE atmospheres with outward-decreasing T show absorption lines.
3. Derive the Milne–Eddington residual-flux profile exactly in LTE.
4. Explain the curve of growth and use the weak-line formula to measure abundances.
5. Recognise when LTE fails for lines, and what NLTE changes.

Key concepts in plain words

A spectral line is a narrow window of high opacity. Atoms absorb strongly only at the frequencies of their transitions. At those frequencies the atmosphere is much more opaque than in the neighbouring continuum.

Why lines have a width. A line is never infinitely narrow, for three reasons: (1) atoms move, so each one sees the light Doppler-shifted (thermal and turbulent motions: a Gaussian core); (2) excited states live only a finite time, so by the uncertainty principle their energy is slightly uncertain (natural broadening); (3) collisions with other particles interrupt the emission (pressure broadening, which grows with density and therefore with surface gravity). The combination is the Voigt profile: a Gaussian core with Lorentzian "damping wings".

Why lines are dark — we see higher, cooler gas. Because the line frequencies are more opaque, at those frequencies we see only down to a higher layer (Eddington–Barbier at $\tau_\nu = 2/3$). In a star whose temperature decreases outward, that layer is cooler and emits less, so the line looks dark. An absorption line in a stellar spectrum is therefore a map of the temperature gradient, not evidence of a separate cold cloud.

Saturation — a line cannot be darker than its top layer. Adding more absorbing atoms makes us see from ever-higher layers, but the source function there approaches a minimum value (the surface value). Once the line core reaches that floor, it cannot get darker: it is *saturated*. Further absorbers only widen the line.

Equivalent width — the "bite" taken out of the spectrum. The equivalent width is the width of a perfectly black rectangle that removes the same amount of light as the line. It measures the total strength of the line, independent of the instrument's resolution, like measuring the area of a bite taken from a biscuit rather than its exact shape.

Curve of growth — from counting atoms to filling the core. For weak lines, each atom removes light independently, so the equivalent width is proportional to the number of absorbers: ideal for measuring abundances. When the core saturates, the width grows very slowly (the flat part). When there are enough atoms to make the damping wings opaque, the width grows again as the square root of the number of atoms.

When LTE fails for lines. In the thin upper layers, collisions are too rare to keep the atomic levels at the local temperature. Line photons escape, the radiation field in the line drops, and the line source function falls below B. Strong-line cores are then *darker* than LTE predicts. Departure coefficients $b_i = n_i / n_i^{\mathrm{LTE}}$ measure how far each level is from its LTE population.

6.1 Line opacity

For a bound–bound transition $l \to u$ with oscillator strength f_{lu}:

$$\alpha_\nu^{\mathrm{line}} = \frac{\pi e^2}{4\pi\varepsilon_0 m_e c}\, f_{lu}\, n_l\, \phi(\nu) \left(1 - e^{-h\nu/kT}\right) = \pi r_e c\, f_{lu}\, n_l\, \phi(\nu) \left(1 - e^{-h\nu/kT}\right) \quad [\mathrm{LTE}],$$

with $r_e = 2.818 \times 10^{-15}$ m, so $\pi r_e c = 2.654 \times 10^{-6}$ m^2 Hz. The level population n_l comes from **Saha–Boltzmann** at the local T and P_e (Lecture SA1). The factor $\left(1 - e^{-h\nu/kT}\right)$ is the LTE stimulated-emission correction.

Profiles ($\int \phi\, d\nu = 1$):

Broadening	Profile	Width
thermal + microturbulent Doppler	Gaussian: $\phi = \dfrac{1}{\sqrt{\pi}\Delta\nu_D} e^{-v^2}$, $v = (\nu - \nu_0)/\Delta\nu_D$	$\Delta\nu_D = \dfrac{\nu_0}{c}\sqrt{\dfrac{2kT}{m} + \xi^2}$
natural, collisional (pressure), Stark	Lorentzian: $\phi = \dfrac{\gamma/4\pi^2}{(\nu - \nu_0)^2 + (\gamma/4\pi)^2}$	$\gamma = \gamma_{\mathrm{rad}} + \gamma_{\mathrm{coll}}$
both	**Voigt**: $\phi = \dfrac{H(a,v)}{\sqrt{\pi}\Delta\nu_D}$, $a = \dfrac{\gamma}{4\pi\Delta\nu_D}$	Gaussian core + Lorentzian ("damping") wings

6.2 Why lines are dark: LTE and Eddington–Barbier

In LTE with pure absorption, both the continuum and the line emit thermally:

$$S_\nu = \frac{\kappa_c B_\nu + \kappa_l B_\nu}{\kappa_c + \kappa_l} = B_\nu(T).$$

The difference lies only in the **depth scale**. With $\eta_\nu \equiv \kappa_l \phi / \kappa_c$, $d\tau_\nu = (1 + \eta_\nu) d\tau_c$. By Eddington–Barbier, $F_\nu \approx \pi B_\nu\big(T(\tau_\nu = 2/3)\big)$:

- In the **continuum**, $\tau_\nu = 2/3$ at $\tau_c = 2/3$: deep and hot.
- At **line centre**, $\tau_\nu = 2/3$ at $\tau_c = \frac{2/3}{1+\eta}$: high and cool.

So $F_{\mathrm{line}} < F_{\mathrm{cont}}$ when T decreases outward. This is the continuous version of MIT 8.901's two-layer toy star (Lecture RT5, §5.4), where the continuum shows $B_\nu(T_H)$ and the lines show $B_\nu(T_C)$. **An LTE absorption line is a map of the temperature gradient.** If T increases outward (a chromosphere), the same physics gives emission lines.

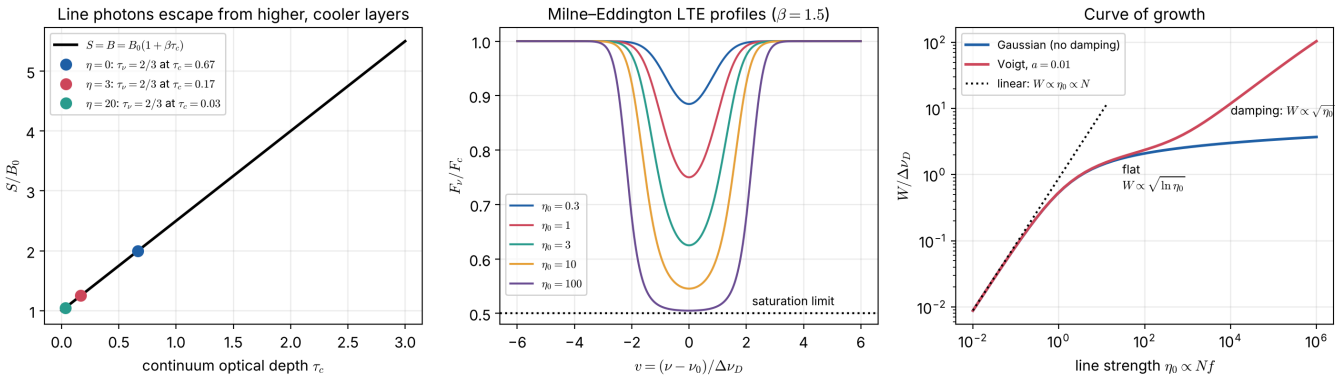


Figure 6.1 — Left: line photons escape from higher layers, where $S = B$ is smaller. Centre: Milne–Eddington LTE profiles for increasing line strength η_0. The core saturates at a finite depth (dotted line), set by the surface value of B. Right: the curve of growth. The equivalent width grows linearly, then flattens (saturation), then rises again as $\sqrt{\eta_0}$ when damping wings become optically thick.

6.3 The Milne–Eddington model: an exact LTE line solution

Assumptions: LTE ($S_\nu = B_\nu$); B_ν linear in continuum optical depth, $B_\nu = B_0(1 + \beta\tau_c)$; and η_ν = constant with depth (the same ratio of line to continuum opacity at all depths).

Then $\tau_\nu = (1 + \eta_\nu)\tau_c$, so $B_\nu = B_0\left(1 + \dfrac{\beta}{1+\eta_\nu}\tau_\nu\right)$ is linear in τ_ν, and Eddington–Barbier is **exact**:

$$F_\nu = \pi B_0\left(1 + \frac{2}{3}\frac{\beta}{1+\eta_\nu}\right), \qquad F_c = \pi B_0\left(1 + \frac{2\beta}{3}\right),$$

$$\boxed{r_\nu \equiv \frac{F_\nu}{F_c} = \frac{1 + \dfrac{2\beta}{3(1+\eta_\nu)}}{1 + \dfrac{2\beta}{3}}}$$

- Weak line ($\eta \ll 1$): $1 - r_\nu \approx \dfrac{2\beta/3}{1 + 2\beta/3}\,\eta_\nu \propto \eta_\nu \propto Nf$.
- Saturated core ($\eta \to \infty$): $r \to \dfrac{1}{1 + 2\beta/3}$. The maximum depth is set by the temperature gradient β. A steeper gradient gives deeper lines; an isothermal atmosphere ($\beta = 0$) gives **no lines at all** in LTE.
- For the gray atmosphere, bolometrically $B \propto \tau + 2/3$, i.e. $\beta = 3/2$, and the maximum depth is $1 - 1/2 = 0.5$. At optical wavelengths β is larger (Wien sensitivity), so lines can be deeper.

6.4 Equivalent width and the curve of growth

The **equivalent width** is the width of a rectangle of zero flux with the same area as the line:

$$W_\nu = \int \left(1 - \frac{F_\nu}{F_c}\right) d\nu, \qquad W_\lambda = \frac{\lambda^2}{c} W_\nu.$$

Weak-line (optically thin) limit. For an absorbing column N_l in front of a continuum (the slab of Lecture RT5 with $S \ll I_0$), $1 - F/F_c \approx \tau_\nu$, so

$$\boxed{W_\lambda = \pi r_e\, \lambda^2\, N_l f} \qquad \text{(independent of the profile and of the broadening)}$$

In SI: $W_\lambda[\mathrm{m}] = 8.853 \times 10^{-15}\, \lambda^2[\mathrm{m}^2]\, N_l f\ [\mathrm{m}^{-2}]$.

Three regimes:

Regime	Condition	W depends on	Use
linear	$\tau_0 \lesssim 1$	$W \propto N f$	**abundances** (best)
flat (saturated)	$1 \lesssim \tau_0 \lesssim 10^{3\text{–}4}$	$W \propto \Delta\nu_D \sqrt{\ln \tau_0}$	**microturbulence** ξ, temperature
damping (square-root)	wings opaque	$W \propto \sqrt{N f \gamma}$	gravity (pressure broadening)

6.5 Beyond LTE

From Lecture SA1, the line source function of a two-level atom is

$$S^{\text{line}} = (1-\varepsilon)\bar{J} + \varepsilon B, \qquad \varepsilon \approx \frac{C_{21}}{C_{21} + A_{21}}.$$

For strong lines in low-density layers, $\varepsilon \ll 1$. Then, from Lecture RT8:

- $S^{\text{line}}(0) \approx \sqrt{\varepsilon}\, B$, well below B, so **NLTE line cores are darker than LTE cores** (e.g. Na I D, Mg I b, Ca II H & K, Hα cores in the Sun).
- Thermalisation occurs only at $\tau_{\text{line}} \sim 1/\sqrt{\varepsilon}$, which can be deep in the photosphere.

Departure coefficients $b_i = n_i / n_i^{\text{LTE}}$ give

$$S^{\text{line}} = \frac{2h\nu^3}{c^2} \frac{1}{\dfrac{b_l}{b_u} e^{h\nu/kT} - 1}, \qquad S^{\text{line}} \approx \frac{b_u}{b_l} B_\nu \quad \text{(Wien)}.$$

When to expect NLTE: low density (giants, supergiants, upper atmospheres); hot stars with intense UV radiation (photo-ionisation dominates); low metallicity (fewer free electrons, so fewer collisions); strong resonance lines; nebulae and the ISM (always NLTE).

	LTE	NLTE
populations	Saha–Boltzmann(T, n_e)	rate equations, coupled to J_ν
S^{line}	$B_\nu(T)$	$(1-\varepsilon)\bar{J} + \varepsilon B$, or via b_i
locality	local	non-local (photons couple distant layers)
computational cost	small	large (ALI, multi-level, many frequencies)
where valid	deep photospheres, weak lines, high density	strong lines, chromospheres, hot or low-density stars

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Worked example 6.1 — Doppler widths

Compute the thermal Doppler width for (a) Hα (656.28 nm) at 5772 K, (b) an Fe I line at 500 nm at 5772 K.

a. $v_{\rm th} = \sqrt{2kT/m_H} = 9.76$ km s^{-1}, so $\Delta\lambda_D = \lambda v/c = 0.0214$ nm $= 214$ mÅ.
b. $m = 55.85\,m_u$: $v_{\rm th} = 1.31$ km s^{-1}, so $\Delta\lambda_D = 21.9$ mÅ. Iron lines are about 10 times narrower than hydrogen lines. In the Sun the microturbulence $\xi \approx 1$ km s^{-1} is comparable to $v_{\rm th}$(Fe), so it matters for metal lines.

Worked example 6.2 — An abundance from a weak line

A weak Fe I line at 500 nm has $W_\lambda = 22$ mÅ and $f = 0.1$. Assume it is on the linear part of the curve of growth and treat the atmosphere as an absorbing column. Find the column density N_l of absorbing atoms.

$$N_l = \frac{W_\lambda}{\pi r_e \lambda^2 f} = \frac{2.2\times10^{-12}}{8.853\times10^{-15}\times(5\times10^{-7})^2\times0.1} = 1.0\times10^{16}\ \mathrm{m}^{-2}.$$

The total iron abundance follows from Saha–Boltzmann (fraction of Fe in the lower level of Fe I) and the column of hydrogen above $\tau \approx 2/3$, roughly $P/(g\mu m_H)$ from Lecture SA5. **LTE enters at the Saha–Boltzmann step.**

Worked example 6.3 — Milne–Eddington line depths

With $\beta = 1.5$, find the central residual flux for $\eta_0 = 1$, 10 and ∞.

$r = \dfrac{1 + 1/(1+\eta)}{2}$: for $\eta = 1$, $r = 0.75$; for $\eta = 10$, $r = 0.545$; for $\eta \to \infty$, $r = 0.5$. Increasing η from 10 to ∞ barely changes the core: the line is **saturated**, and further growth of W happens only in the wings.

Worked example 6.4 — LTE emission lines from a chromosphere

Suppose that above the temperature minimum T rises outward, so that near the surface $B = B_0(1 - \beta'\tau_c)$ with $\beta' > 0$ for $\tau_c < 0.1$. What does LTE predict for a strong line formed at $\tau_c \approx 0.01$?

The strong line's $\tau_\nu = 2/3$ falls at $\tau_c \approx 0.01$, where B is larger than at $\tau_c \approx 0.1$ (the temperature minimum), so the core shows an **emission reversal** inside the broad absorption profile. This is seen in the Ca II H & K lines (the K_2 peaks). In reality NLTE decoupling ($S < B$) makes the emission weaker than LTE predicts.

Common misconceptions

1. **"Line depth is proportional to abundance."** Only for weak lines; saturated lines barely respond to abundance changes.
2. **"Absorption lines in a stellar spectrum come from a cold gas layer in front of the star."** They come from the outward decrease of temperature in the atmosphere itself.
3. **"LTE always overestimates line strengths."** For strong resonance lines LTE usually predicts cores that are too *bright*; NLTE makes them darker.
4. **"Line width only measures temperature."** It also includes microturbulence, rotation, pressure broadening (gravity) and natural damping.

Lecture summary

- Line opacity uses Saha–Boltzmann populations (LTE) and a Voigt profile.
- Lines are dark because, at line frequencies, $\tau_\nu = 2/3$ lies in higher, cooler layers.
- Milne–Eddington: $r = [1 + 2\beta/3(1+\eta)]/(1 + 2\beta/3)$; saturation depth is set by β.
- Curve of growth: linear ($W \propto Nf$, abundances) → flat → square-root (damping).
- NLTE: $S^{\text{line}} = (1-\varepsilon)\bar{J} + \varepsilon B$; strong-line cores are darker than in LTE.

Exam questions and answers — Lecture SA6

Q1. Explain, using the Eddington–Barbier relation, why the solar spectrum shows absorption lines. What would an LTE isothermal atmosphere show?

> **A1.** $F_\nu \approx \pi S_\nu(\tau_\nu = 2/3)$, and in LTE $S_\nu = B_\nu(T)$. Line opacity raises τ_ν, so $\tau_\nu = 2/3$ is reached higher up, where T (hence B_ν) is lower, and the flux is lower. If the atmosphere were isothermal, B_ν would be the same at all depths and $F_\nu = F_c$: no lines, whatever the opacity.

Q2. Derive the Milne–Eddington residual flux and its saturation limit.

> **A2.** See §10.3. $B = B_0(1 + \beta\tau_c)$ and $\tau_\nu = (1+\eta)\tau_c$ give $B = B_0(1 + \beta\tau_\nu/(1+\eta))$, which is linear in τ_ν. Then $F = \pi B(\tau_\nu = 2/3) = \pi B_0(1 + \frac{2\beta}{3(1+\eta)})$. Divide by F_c ($\eta = 0$). As $\eta \to \infty$, $r \to 1/(1 + 2\beta/3)$.

Q3. Sketch the curve of growth and explain the physics of each of its three parts. Which part is best for abundance determination and why?

> **A3.** Linear: the line is optically thin; every atom removes photons independently, so $W \propto Nf$. Flat: the core is saturated ($r \to r_{\min}$), and W grows only as the Doppler core broadens in frequency where $\tau_\nu > 1$, i.e. $W \propto \Delta\nu_D\sqrt{\ln \tau_0}$. Square-root: the Lorentzian damping wings ($\propto \gamma/\Delta\nu^2$) become optically thick, giving $W \propto \sqrt{Nf\gamma}$. The linear part is best for abundances because W is insensitive to the poorly known broadening parameters (ξ, γ).

Q4. Why are the cores of strong resonance lines in the Sun deeper than LTE models predict?

> **A4.** In the low-density upper photosphere and chromosphere, $C_{21} \ll A_{21}$, so $\varepsilon \ll 1$ and $S^{\text{line}} = (1-\varepsilon)\bar{J} + \varepsilon B$. Photons escape from the line core ($\bar{J} < B$) and the source function drops to $\sim \sqrt{\varepsilon}\, B$ at the surface. The core intensity $\approx S(\tau_\nu \approx 1)$ is therefore lower than $B(T)$, and the line is darker than in LTE.

Lecture SA7 — TE, LTE and Non-LTE: Definitions, Examples and Solution Methods

> **Sources:** Collins §9.1 (breakdown of strict TE; assumption of LTE); Chandrasekhar (1950) Ch. I; R. N. Thomas, *Some Aspects of Non-Equilibrium Thermodynamics in the Presence of a Radiation Field* (1965); Mihalas, *Stellar Atmospheres* (1978) Chs. 5, 11–12; Hubeny & Mihalas, *Theory of Stellar Atmospheres* (2014); Osterbrock & Ferland (2006) for nebulae. This lecture extends Lecture SA1 (LTE) and Lecture RT8 (scattering).

Learning goals

1. Give precise definitions of **thermodynamic equilibrium (TE)**, **local thermodynamic equilibrium (LTE)** and **non-LTE (NLTE)**, and name astrophysical examples of each.
2. Decide, with a quantitative criterion, which regime applies to a given object.
3. Write the statistical-equilibrium (rate) equations and the NLTE line source function with departure coefficients.
4. Explain the main NLTE mechanisms (photon losses, overionisation, pumping) and their effect on spectra and abundances.
5. Name the main NLTE solution methods and codes (names only).

Key concepts in plain words

Three levels of "equilibrium". Think of a crowded hall. In **TE** the hall is closed and has been quiet for a long time: everyone, people (matter) and the air in the room (radiation), shares the same temperature everywhere. In **LTE** the doors are open and air flows through (radiation streams out), but people still bump into each other so often that each corner of the hall keeps its own well-defined local temperature. In **NLTE** the hall is so empty that people rarely meet; how they behave is decided by what blows in through the doors (the radiation field), not by the local temperature.

What decides the regime: who talks to the atoms? An atom's levels are populated and emptied by two kinds of events: collisions (with electrons and atoms, which "know" the local temperature) and photons (which may come from far away, at a different temperature, or may escape altogether). If collisions dominate, the populations follow the local temperature: LTE. If photons dominate, the populations follow the radiation field: NLTE.

NLTE is the rule in low-density gas. Most of the Universe by volume is NLTE: nebulae, the interstellar medium, stellar chromospheres and coronae, winds, the upper atmospheres of planets. LTE is a very good approximation mainly in dense places such as the deep photospheres of stars and stellar interiors.

Why it matters for measurements. If you assume LTE when a line actually forms in NLTE, you misread its strength. You then derive wrong abundances (sometimes by a factor of 2 or more), wrong temperatures and wrong gravities. This is why modern abundance work quotes "NLTE corrections".

7.1 Definitions

Thermodynamic equilibrium (TE)

> **Definition.** A system is in **TE** at temperature T when *every* process is balanced by its inverse (detailed balance) and all distributions take their equilibrium form at the **same** T: Maxwell velocities, Boltzmann excitation, Saha ionisation, and Planck radiation with $I_\nu = B_\nu(T)$, isotropic. Consequently $J_\nu = B_\nu$, $F_\nu = 0$ and $S_\nu = B_\nu$.

Examples of TE (or extremely close to it)

- A closed oven (Hohlraum) with a tiny hole: the classical black body.
- The **deep interior of the Sun**: the photon mean free path ($\sim$ cm) is 10^{-10}–10^{-11} of the temperature scale height, so the radiation is isotropic to that precision (Worked example 7.1).
- The **early Universe before recombination** ($z \gtrsim 1100$): matter and radiation were coupled by Thomson scattering at one temperature. This is why the CMB has a Planck spectrum.

Local thermodynamic equilibrium (LTE)

> **Definition. LTE** holds when the **matter** at each point is in equilibrium at the **local** temperature $T(\mathbf{r})$ (Maxwell, Boltzmann, Saha at $T(\mathbf{r})$), while the **radiation** is free to differ from B_ν. Then Kirchhoff's law $j_\nu = \kappa_\nu \rho B_\nu(T)$ holds and $S_\nu = B_\nu(T)$, but in general $I_\nu \neq B_\nu$ and $J_\nu \neq B_\nu$.

Examples of LTE (good approximation)

- The **continuum** of the solar photosphere (H^- opacity) and **weak metal lines** in dwarf F, G, K stars: the basis of classical abundance analysis.
- The **gray model atmosphere** of Lecture SA5 and the Milne–Eddington lines of Lecture SA6.
- **Earth's lower atmosphere** in the thermal infrared: the CO_2 15 μm band is in LTE below about 70 km, which is why climate radiative-transfer codes use LTE there.
- Dense laboratory plasmas (arcs, shock tubes).

Non-LTE (NLTE)

> **Definition. NLTE** means that at least some level populations differ from their Saha–Boltzmann values at the local T. The populations must then be computed from the **statistical-equilibrium (rate) equations**, which couple them to the radiation field. Then $S_\nu \neq B_\nu(T)$, and the problem becomes non-local and non-linear.

Examples of NLTE

- **Strong resonance-line cores** in the Sun (Na I D, Mg I b, Ca II H & K, Hα core): photons escape, so $S < B$ and the cores are darker than LTE predicts.
- The **solar chromosphere and corona**: low density; populations controlled by radiation and by non-thermal heating.

- **Hot O and B stars and their winds**: intense UV radiation drives photo-ionisation; He II and N III lines need NLTE. **Supergiants** have low gravity and therefore low density.
- **H II regions and planetary nebulae**: forbidden lines such as [O III] 500.7 nm, with electron densities far below the critical density (Worked example 7.3).
- **Molecular clouds**: high rotational lines of CO are "subthermally" excited ($T_{\rm ex} < T_{\rm kin}$).
- **Masers** (OH, H_2O, SiO): extreme NLTE with population inversion, $T_{\rm ex} < 0$.
- **Earth's upper atmosphere**: aurorae, airglow, and CO_2 cooling above about 70 km.

Side-by-side comparison

	TE	LTE	NLTE
temperature	one T everywhere	local $T(\mathbf{r})$	kinetic $T(\mathbf{r})$, but populations not set by it
velocities	Maxwell(T)	Maxwell($T(\mathbf{r})$)	usually Maxwell($T(\mathbf{r})$)
excitation, ionisation	Boltzmann, Saha(T)	Boltzmann, Saha($T(\mathbf{r})$)	rate equations: $n_i = b_i n_i^{\rm LTE}$
source function	$S_\nu = B_\nu$	$S_\nu = B_\nu(T(\mathbf{r}))$	$S_\nu \neq B_\nu$, depends on J_ν
radiation field	$I_\nu = B_\nu$, isotropic, $F = 0$	$I_\nu \neq B_\nu$, carries flux	$I_\nu \neq B_\nu$
local or global?	trivial	local: $T \to$ populations $\to$ κ_ν, j_ν	global: populations $\leftrightarrow$ radiation everywhere
criterion	closed and isothermal over $\gg \ell_\gamma$	collision rates $\gg$ radiative rates	radiative rates $\gtrsim$ collision rates
examples	oven, stellar core, early Universe	photospheric continuum, weak lines, Earth's troposphere	resonance-line cores, chromospheres, hot stars, nebulae, masers

A practical criterion

For a transition $u \to l$, collisions win if

$$n_e\, q_{ul} \;\gg\; A_{ul}\, \beta_{\rm esc},$$

where q_{ul} is the collisional de-excitation rate coefficient and $\beta_{\rm esc} \approx 1/\tau$ (for $\tau \gg 1$) is the probability that an emitted photon escapes. In words: compare the **critical density** $n_{\rm crit} = A_{ul}\beta_{\rm esc}/q_{ul}$ with the actual electron density. Optically thick lines trap their photons ($\beta_{\rm esc} \ll 1$), which lowers the effective critical density and pushes them back toward LTE deep inside an atmosphere. Near the surface, where $\tau \to 0$, that help disappears.

7.2 Statistical equilibrium

For each level i of an atom or ion:

$$\boxed{n_i \sum_{j \neq i} \left(R_{ij} + C_{ij}\right) = \sum_{j \neq i} n_j \left(R_{ji} + C_{ji}\right)} \qquad (\text{rates out} = \text{rates in})$$

- **Radiative bound–bound rates:** $R_{lu} = B_{lu}\bar{J}$ (upward); $R_{ul} = A_{ul} + B_{ul}\bar{J}$ (downward).

- **Radiative bound–free rates:** photo-ionisation $R_{i\kappa} = 4\pi \int \frac{\sigma_i(\nu)}{h\nu} J_\nu d\nu$, plus spontaneous and stimulated recombination.
- **Collisional rates:** $C_{ij} = n_e q_{ij}(T)$, obeying detailed balance at the local T.

Because J_ν depends on the populations everywhere through the RTE, and the populations depend on J_ν, the problem is **non-linear and non-local**.

Departure coefficients $b_i = n_i / n_i^{\mathrm{LTE}}$ ($b_i = 1$ in LTE). The line source function is

$$S_\nu^{\mathrm{line}} = \frac{2h\nu^3}{c^2} \frac{1}{\dfrac{b_l}{b_u} e^{h\nu/kT} - 1} \approx \frac{b_u}{b_l} B_\nu(T) \quad (\text{Wien limit}).$$

For a two-level atom this becomes $S = (1-\varepsilon)\bar{J} + \varepsilon B$ (Lecture SA1).

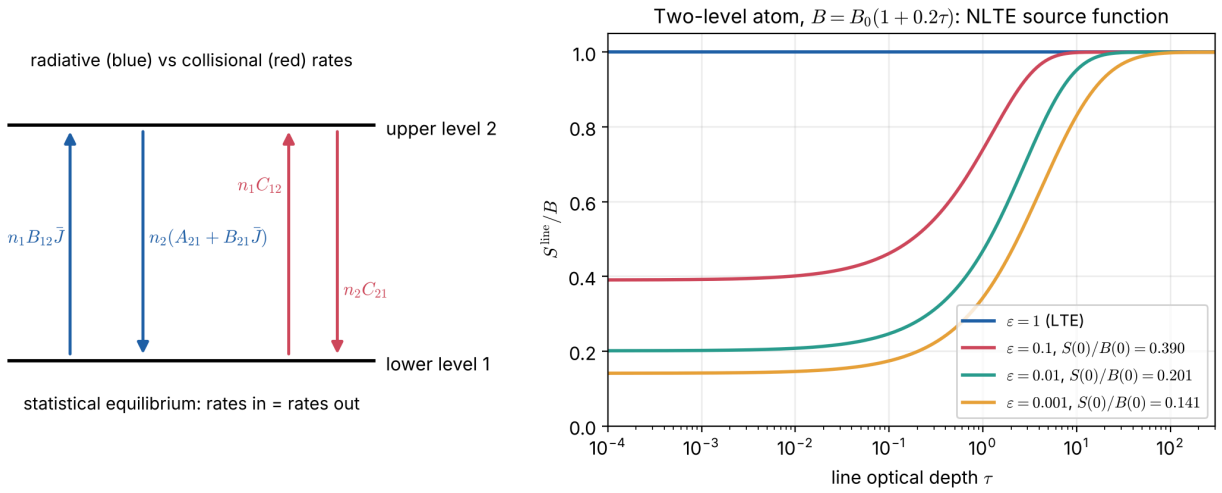


Figure 7.1 — Left: the four rate processes that set the populations of a two-level atom. Right: the exact line source function (Λ-matrix solution with `scripts/lte_solvers.py`) in an atmosphere whose Planck function rises gently inward. As collisions become rarer ($\varepsilon \downarrow$), the surface source function falls far below B, and thermalisation ($S \to B$) happens deeper, at $\tau \sim 1/\sqrt{\varepsilon}$.

7.3 NLTE mechanisms and their effects

Mechanism	What happens	Effect on spectra	Typical cases
photon losses (resonance scattering)	line photons escape near the surface, so $\bar{J} < B$, $b_u < b_l$ and $S < B$	deeper line cores than LTE	Na D, Ca II H & K, Hα core, Mg II h & k
overionisation	UV $J_\nu > B_\nu$ ionises minority species	weaker lines → LTE underestimates abundance (positive NLTE correction)	Fe I, Mg I in metal-poor stars
overrecombination	recombination cascades overpopulate upper levels	emission features	He I, H I in nebulae and winds
photon pumping / fluorescence	a strong radiation field populates particular upper levels	emission lines, line-strength changes	Bowen fluorescence (O III), masers
photon suction	cascades drain a level through strong lines	modified populations	high-excitation lines in the solar spectrum

Typical **NLTE abundance corrections** $\Delta_{\rm NLTE} = \log \varepsilon_{\rm NLTE} - \log \varepsilon_{\rm LTE}$: about −0.2 dex for the O I 777 nm triplet in the Sun, and up to a few tenths of a dex for Fe I and Mg I in metal-poor giants.

7.4 NLTE solution methods: names only

Formal solvers (for a given source function):
Feautrier method · long characteristics · short characteristics (linear, parabolic, Bézier/monotonic) · DELO and Hermitian methods · discontinuous finite elements · discrete ordinates · Monte Carlo radiative transfer.

Iterative schemes for scattering and statistical equilibrium:
classical Λ-iteration · core saturation (Rybicki) · operator perturbation / accelerated Λ-iteration (ALI) with diagonal, tridiagonal or nonlocal approximate operators · multilevel ALI (MALI) · Gauss–Seidel and successive over-relaxation (GS/SOR) · multigrid · Ng acceleration · Krylov-subspace methods (GMRES, BiCGStab) with preconditioning · complete linearization · hybrid complete-linearization/ALI (CL/ALI) · variable Eddington factors · equivalent two-level atom · escape-probability and Sobolev methods · comoving-frame methods · partial-redistribution (PRD) schemes.

Codes (examples):
MULTI, MULTI3D, RH, Lightweaver, DETAIL/SURFACE, TLUSTY/SYNSPEC, PHOENIX, CMFGEN, FASTWIND, PoWR, Balder, Cloudy, MOCASSIN, LIME, RADEX.

Worked example 7.1 — How close to TE is the centre of the Sun?

With a photon mean free path $\ell \approx 2$ cm (Lecture RT4) and a temperature scale height $H_T = T/|dT/dr| \approx R_\odot \approx 7 \times 10^8$ m, estimate the anisotropy of the radiation, $F/(cu)$.

Diffusion gives $F = \frac{c\ell}{3}|\nabla u|$ with $u \propto T^4$, so $|\nabla u|/u = 4/H_T$. Then $F/(cu) = \frac{4\ell}{3H_T} = \frac{4\times 0.02}{3\times 7\times 10^8} \approx 4 \times 10^{-11}$. The radiation is isotropic and Planckian to about 1 part in 10^{11}, essentially TE, although this tiny anisotropy carries the entire solar luminosity.

Worked example 7.2 — Is the Na D line in LTE?

Na I D_2 has $A_{21} = 6.2 \times 10^7$ s^{-1}; electron collisions give $q_{21} \approx 6 \times 10^{-13}$ m^3 s^{-1} (order of magnitude, $T \approx 5000$ K). Compare with $n_e \approx 6 \times 10^{19}$ m^{-3} at $\tau_{500} = 1$ and $n_e \approx 10^{17}$ m^{-3} in the upper photosphere where the line core forms.

$n_{\text{crit}} = A/q \approx 10^{20}$ m^{-3}. At $\tau_{500} = 1$: $C/A = n_e q/A \approx 0.6$, close to LTE only if line trapping helps. Where the core forms: $C/A \approx 10^{-3}$, so $\varepsilon \approx 10^{-3}$, strongly NLTE. In the model of Figure 7.1, $\varepsilon = 10^{-3}$ gives $S(0)/B(0) \approx 0.14$. The observed Na D cores are indeed much darker than LTE models predict.

Worked example 7.3 — A planetary nebula

[O III] 500.7 nm has a critical density of about 6×10^{11} m^{-3} (6×10^5 cm^{-3}). A planetary nebula has $n_e \approx 10^{10}$ m^{-3}. Which regime applies, and what does this make possible?

$n_e/n_{\text{crit}} \approx 0.016$: every collisional excitation is followed by a radiative decay rather than a collisional de-excitation, which is extreme NLTE. The line emissivity is then $\propto n_e n_{\text{ion}} q_{lu}(T_e)$, and the ratio of lines from different upper levels ([O III] 436.3 / 500.7 nm) depends only on T_e. This is how nebular electron temperatures (about 10 000 K) are measured. In LTE such a ratio would just reflect $e^{-\Delta E/kT}$, and the forbidden lines would be very weak.

Worked example 7.4 — The CMB: TE in the past, a Planck spectrum today

After recombination ($z \approx 1100$) photons stopped interacting with matter. Why is the CMB still a perfect Planck spectrum today, at $T = 2.725$ K?

Before recombination matter and radiation were in TE at $T \approx 3000$ K. Afterwards the photons stream freely: I_ν/ν^3 is conserved along each ray (Lecture RT2), and every frequency redshifts by the same factor $1 + z$. A Planck spectrum at T becomes a Planck spectrum at $T/(1+z)$, so $3000/1100 \approx 2.7$ K. Today's CMB is **not** in TE with any matter (the radiation is decoupled), yet its spectrum is still exactly Planckian. A Planck spectrum alone does not prove equilibrium at present.

Common misconceptions

1. **"NLTE means the gas has no temperature."** The electrons and atoms usually still have a Maxwellian distribution with a well-defined kinetic T; only the level populations (and therefore S_ν) depart from their LTE values.
2. **"A Planck spectrum proves TE."** The CMB today is Planckian but decoupled; conversely, LTE matter can emit non-Planckian spectra.
3. **"NLTE always weakens lines."** Photon losses deepen line cores; overionisation weakens lines of minority species; pumping can create emission. The sign depends on the mechanism.
4. **"High density guarantees LTE everywhere in a star."** Near the surface photons escape ($\beta_{\text{esc}} \to 1$), so even dense atmospheres show NLTE in strong lines.

Lecture summary

- **TE:** one T for everything, $I_\nu = B_\nu$. Examples: oven, stellar core, early Universe.

- **LTE:** matter at local T, so $S_\nu = B_\nu(T)$, but $I_\nu \neq B_\nu$. Examples: photospheric continuum, weak lines, Earth's troposphere.
- **NLTE:** populations from rate equations, $S_\nu \neq B_\nu$. Examples: resonance-line cores, chromospheres, hot stars, nebulae, masers.
- Criterion: compare n_e with $n_{\rm crit} = A\beta_{\rm esc}/q$.
- Mechanisms: photon losses, overionisation, overrecombination, pumping. Methods: Λ-iteration, ALI/MALI, complete linearization, CL/ALI, GS/SOR, Monte Carlo, among others.

Exam questions and answers — Lecture SA7

Q1. Define TE, LTE and NLTE precisely and give two astrophysical examples of each.

> **A1. TE:** all distributions (Maxwell, Boltzmann, Saha, Planck) at one temperature, with detailed balance for every process and $I_\nu = B_\nu$; examples: the solar core, the pre-recombination Universe. **LTE:** matter in equilibrium at the local temperature, $S_\nu = B_\nu(T(\mathbf{r}))$, but the radiation can differ from B_ν; examples: the solar photospheric continuum, weak Fe lines in the Sun (or Earth's troposphere in the infrared). **NLTE:** some populations differ from Saha–Boltzmann and must be computed from rate equations coupled to the radiation; examples: the cores of Na D and Ca II H & K, planetary nebulae (forbidden lines), astrophysical masers.

Q2. Write the statistical-equilibrium equation for level i and explain why NLTE problems are non-local and non-linear.

> **A2.** $n_i \sum_j (R_{ij} + C_{ij}) = \sum_j n_j (R_{ji} + C_{ji})$. The radiative rates R depend on J_ν, which by the RTE depends on the emissivities and opacities, i.e. on the populations, at all points of the atmosphere (non-local). The opacities are products of populations and radiation-dependent quantities, and the populations depend on J (non-linear). The two sets of equations must be solved together iteratively.

Q3. Why are the cores of strong resonance lines darker in NLTE than in LTE? Use departure coefficients.

> **A3.** Near the surface the line photons escape, so $\bar{J} < B$ and fewer atoms are radiatively excited: the upper level is underpopulated relative to the lower, $b_u < b_l$. Then $S^{\rm line} \approx (b_u/b_l)B < B$. The emergent core intensity ($\approx S$ at $\tau_\nu \approx 1$, Eddington–Barbier) is therefore lower than the LTE value $B(T)$, and the core is darker.

Q4. Name four methods used to solve NLTE radiative-transfer problems and two NLTE codes.

> **A4.** Methods (any four): Λ-iteration, accelerated Λ-iteration (ALI), multilevel ALI (MALI), complete linearization, hybrid CL/ALI, Gauss–Seidel/SOR, Feautrier formal solution, short characteristics, Monte Carlo, escape-probability/Sobolev methods. Codes (any two): MULTI, RH, TLUSTY, PHOENIX, CMFGEN, FASTWIND, DETAIL/SURFACE, Cloudy.

Lecture SA8 — Stellar Atmospheres: From the Gray Model to Real Stars

> **Sources:** Collins Part II, Ch. 9 (assumptions of normal stellar atmospheres §9.1, the role of the atmosphere in setting boundary conditions); Gray, *The Observation and Analysis of Stellar Photospheres* (2005); Mihalas, *Stellar Atmospheres* (1978); Hubeny & Mihalas (2014); Vernazza, Avrett & Loeser (1981) for the VAL-C solar model; A670 Lecture RT1 (applications).

Learning goals

1. List the equations of a classical model atmosphere and the iteration used to solve them.
2. Explain the effects of non-gray opacity: line blanketing, surface cooling and backwarming; know the dominant opacity sources across the spectral sequence.
3. Decide when convection, sphericity, winds or 3D effects must be included.
4. Describe the layers of the solar atmosphere and why the upper layers need non-radiative heating and NLTE.
5. Know how $T_{\rm eff}$, $\log g$ and abundances are measured, and name the main model-atmosphere grids and codes.

Key concepts in plain words

The atmosphere is the star's "skin" and its only messenger. Everything we learn about a star comes through its atmosphere, a layer usually less than 0.1 % of the stellar radius. It sets the observed spectrum, and it also sets the outer boundary condition for the stellar interior.

A model atmosphere is a self-consistent recipe. Give it three numbers (the effective temperature, the surface gravity and the chemical composition) and it computes, depth by depth, the temperature, pressure, density, ionisation, opacity and radiation field. All of these must agree with each other: the energy must flow outward at a constant rate, gas pressure must balance gravity, and the populations must follow from T and P (in LTE) or from the rate equations (in NLTE).

Real opacity is not gray. Thousands of spectral lines block radiation in some frequency windows. The energy is pushed into other windows and back into the deeper layers, which heat up ("backwarming"), while the upper layers cool. This line blanketing changes colours, temperature structures and derived abundances.

Beyond the classical picture. Cool stars have convection reaching into the photosphere, visible as granulation. Giants and supergiants have extended atmospheres where the plane-parallel approximation fails. Hot stars lose mass in radiation-driven winds. The Sun has a chromosphere and a million-degree corona heated by non-radiative energy. Each of these requires physics beyond the gray LTE model of Lecture SA5, and modern models handle them with spherical, NLTE, time-dependent or 3D methods.

8.1 The equations of a classical model atmosphere

Assumptions (Collins §9.1d, extended): plane-parallel, static, 1D, horizontally homogeneous, LTE (or NLTE), energy carried by radiation and convection, no energy sources.

Equation	Form	Role
hydrostatic equilibrium	$\frac{dP_{\rm gas}}{d\tau} = \frac{g}{\kappa} - \frac{dP_{\rm rad}}{d\tau}$ $(-\ \text{turbulent pressure})$	pressure structure
energy conservation	$F_{\rm rad} + F_{\rm conv} = \sigma T_{\rm eff}^4$	temperature structure
radiative transfer	$\mu\, dI_\nu/d\tau_\nu = I_\nu - S_\nu$ at about 10^4–10^6 frequencies	radiation field
convection	mixing-length theory when $\nabla_{\rm rad} > \nabla_{\rm ad}$	$F_{\rm conv}$
equation of state	ideal gas + Saha–Boltzmann (LTE) or rate equations (NLTE)	populations, n_e, ρ
opacity	$\kappa_\nu(T, P, \text{composition})$: continuum + lines + molecules	couples everything

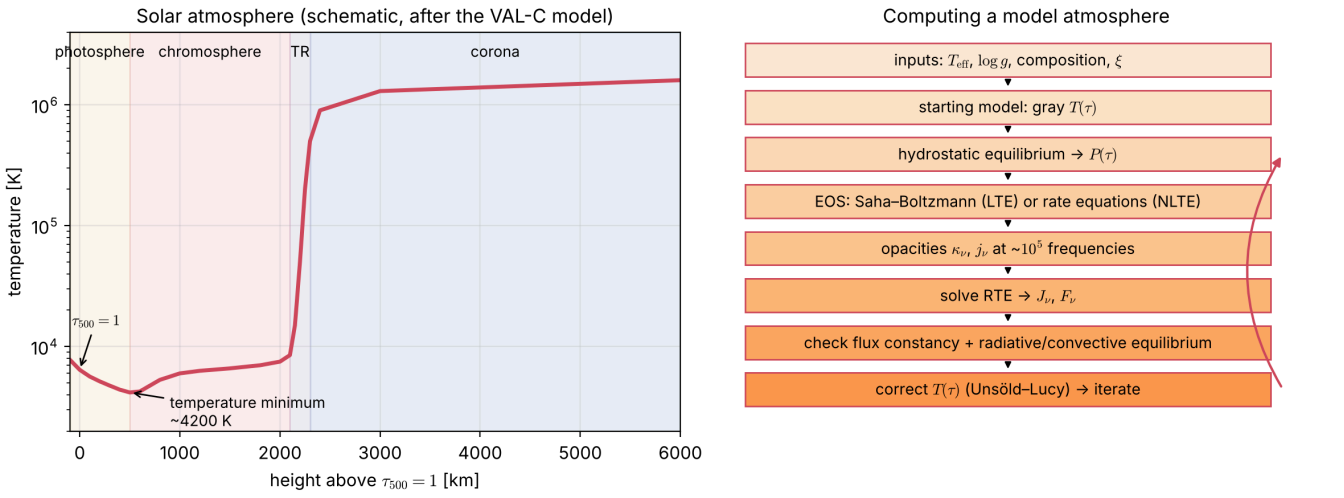


Figure 8.1 — Left: the layers of the solar atmosphere (schematic, after the VAL-C model of Vernazza, Avrett & Loeser 1981): the temperature falls through the photosphere to a minimum near 4200 K, rises in the chromosphere, jumps through the thin transition region (TR), and reaches 1–2 MK in the corona. Right: how a model atmosphere is computed. Start from the gray $T(\tau)$ of Lecture SA5 and iterate until the flux is constant at every depth.

Temperature correction. At each iteration the error in flux constancy is used to correct $T(\tau)$. The classical scheme is the **Unsöld–Lucy** correction; others include the Avrett–Krook method and complete linearization.

8.2 Non-gray effects and opacity sources

Spectral type	$T_{\rm eff}$ [K]	Dominant continuum opacity	Notable spectral features
O, early B	30 000–50 000	electron scattering; He II and He I bound-free	He II lines; winds; NLTE essential
late B, A	10 000–20 000	H I bound-free (Balmer, Paschen continua)	strongest Balmer lines and Balmer jump near A0
F, G, K	4 000–7 500	$\mathbf{H^-}$ bound-free and free-free	metal lines, G band (CH), Ca II H & K
M	2 500–4 000	H^- + molecules (TiO, VO, H_2O)	molecular bands dominate
L, T (brown dwarfs)	< 2 500	molecules (H_2O, CH_4), dust clouds	methane bands, clouds

Line blanketing (blocked flux in line-crowded regions, mainly the UV) has three effects:

1. **Blocking:** the UV flux is reduced, so colours change.
2. **Backwarming:** to carry the same total flux through the blocked layers, the deeper layers become hotter.
3. **Surface cooling:** lines are efficient emitters in the cool upper layers, which lowers the boundary temperature below the gray value.

8.3 Convection in atmospheres

The **Schwarzschild criterion**: convection occurs if $\nabla_{\rm rad} \equiv \left(\dfrac{d\ln T}{d\ln P}\right)_{\rm rad} > \nabla_{\rm ad}$ ($\nabla_{\rm ad} = 0.4$ for a monatomic ideal gas, smaller where hydrogen ionises).

For the gray atmosphere with constant κ ($P \propto \tau, T^4 \propto \tau + \frac{2}{3}$): $\nabla_{\rm rad} = \dfrac{\tau}{4(\tau + 2/3)} < \dfrac{1}{4} < 0.4$, so it is **stable**. In real cool stars κ rises steeply with T (H^-), and hydrogen ionisation lowers $\nabla_{\rm ad}$, so convection starts just below the photosphere. Its top is seen as **granulation** (cells about 1000 km across, lifetimes of about 5–10 min on the Sun). 1D models use mixing-length theory and a “microturbulence” parameter; **3D radiation-hydrodynamic** models (e.g. Stagger, CO5BOLD, MURaM, Bifrost) simulate convection directly.

8.4 The solar atmosphere: layers and physics

Layer	Height [km]	T	Physics	Diagnostics
photosphere	0–500	6400 → 4200 K	radiative equilibrium, mostly LTE, convective overshoot	visible continuum, weak lines, limb darkening
temperature minimum	about 500	about 4200 K		UV continuum near 160 nm, far IR
chromosphere	500–2100	6000–20 000 K	non-radiative (wave and magnetic) heating; NLTE; partial ionisation	Hα, Ca II H & K, Mg II h & k, mm-wave continuum
transition region	about 2100–2300	$10^4 \rightarrow 10^6$ K	steep conduction-dominated gradient	UV lines of C IV, Si IV, O V
corona	> 2300	$1–2 \times 10^6$ K	magnetic heating, collisional ionisation equilibrium (coronal approximation)	EUV and X-ray lines (Fe IX–XVI), white-light K-corona (Thomson scattering)

The rise of T above the temperature minimum **cannot** be produced by radiative equilibrium. It requires mechanical or magnetic heating, and its spectral lines (Hα, Ca II, Mg II) must be modelled in NLTE (Lecture SA7). In LTE, the temperature rise produces emission cores in strong lines, the thermal-inversion effect of the MIT toy model (Lecture RT5).

8.5 Beyond plane-parallel and static

- **Sphericity.** The atmosphere's thickness is roughly $10H$, where $H = kT/(\mu m_H g)$ is the pressure scale height. When $10H/R$ exceeds a few per cent (giants, supergiants), spherical models are needed (e.g. spherical MARCS, PHOENIX).
- **Stellar winds.** Hot stars drive winds by the radiation force on lines (radiation-driven wind theory of Castor, Abbott & Klein, 1975). Lines form in expanding gas and show **P Cygni profiles** (blue-shifted absorption plus emission). The transfer is solved with the **Sobolev approximation** or in the comoving frame (CMFGEN, PoWR, FASTWIND).
- **3D and time dependence.** Granulation, magnetic structures, pulsation and shocks (Miras) need 3D or time-dependent radiation-hydrodynamics. 3D NLTE analyses lowered the solar oxygen abundance by about 0.2 dex in the 2000s (Asplund et al. 2009).

8.6 Model grids and codes

Code or grid	Type	Typical use
ATLAS9, ATLAS12 (Kurucz)	1D, LTE, plane-parallel, line-blanketed	A–K stars; classic grids
MARCS (Gustafsson et al.)	1D, LTE, plane-parallel or spherical, molecules	F–M stars, giants
PHOENIX (Hauschildt & Baron)	1D or 3D, spherical, NLTE capable	cool stars, brown dwarfs, novae, supernovae
TLUSTY (Hubeny & Lanz)	1D NLTE, plane-parallel	O and B stars, white dwarfs, accretion disks (OSTAR, BSTAR grids)
CMFGEN, PoWR, FASTWIND	spherical NLTE with winds	massive stars, Wolf–Rayet stars, supernovae
Stagger, CO5BOLD, MURaM, Bifrost	3D radiation-(magneto)hydrodynamics	granulation, solar atmosphere, abundances
SYNTHE, MOOG, Turbospectrum, SME, SYNSPEC	spectrum synthesis and analysis	abundances, line profiles

8.7 How stellar parameters are measured

Parameter	Main diagnostics
$T_{\rm eff}$	spectral energy distribution and colours; infrared flux method; Balmer-line wings (F–G); excitation balance of Fe I lines; line-depth ratios; interferometric angular diameter + bolometric flux
$\log g$	pressure-broadened wings (Mg I b, Ca II, Balmer lines in hot stars); ionisation balance (Fe I/Fe II); asteroseismology; mass and radius from binaries
abundances	equivalent widths on the linear part of the curve of growth (Lecture SA6); spectrum synthesis with LTE or NLTE and 1D or 3D models
microturbulence ξ	Fe I abundance independent of equivalent width (flat part of the curve of growth)
magnetic field	Zeeman splitting and Stokes V (Lecture RT9)

Worked example 8.1 — When is an atmosphere plane-parallel?

Compute the pressure scale height $H = kT/(\mu m_H g)$ with $\mu = 1.3$ for (a) the Sun ($T = 5772$ K, $g = 274$ m s^{-2}), (b) a red supergiant ($T = 4000$ K, $\log g = 0$ cgs, i.e. $g = 0.01$ m s^{-2}, $R = 500 R_\odot$), (c) a white dwarf ($T = 10^4$ K, $\log g = 8$, $g = 10^6$ m s^{-2}).

a. $H = \dfrac{1.381\times10^{-23}\times 5772}{1.3\times 1.674\times10^{-27}\times 274} = 1.34\times10^{5}$ m = 134 km; $10H/R_\odot = 0.2\,\%$, so plane-parallel is excellent.

b. $H = 2.54 \times 10^9$ m $= 3.6\,R_\odot$; $10H/R = 7\,\%$, so **spherical** models are needed.
c. $H \approx 64$ m: the whole atmosphere is a few hundred metres thick; extremely plane-parallel, with huge pressure broadening.

Worked example 8.2 — Why the gray atmosphere is not convective

Show that $\nabla_{\rm rad} < 0.25$ for the Eddington gray atmosphere with constant opacity, and compare with $\nabla_{\rm ad} = 0.4$.

$P = g\tau/\kappa \Rightarrow d\ln P = d\ln\tau$. With $T^4 = \frac{3}{4}T_{\rm eff}^4(\tau + \frac{2}{3})$: $4\,d\ln T = \dfrac{d\tau}{\tau + 2/3}$, so

$$\nabla_{\rm rad} = \frac{d\ln T}{d\ln \tau} = \frac{\tau}{4(\tau + 2/3)}.$$

At $\tau = 1$: 0.15; as $\tau \to \infty$: 0.25. Always below 0.4, so the gray, constant-κ atmosphere is convectively stable. Real solar-type stars are convective just below the photosphere because κ grows steeply with T and $\nabla_{\rm ad}$ drops in the hydrogen ionisation zone.

Worked example 8.3 — Energy budget of the chromosphere

The solar chromosphere radiates about 4×10^3 W m^{-2} (mainly in Hα, Ca II and Mg II), and the corona about 3×10^2 W m^{-2} in the quiet Sun. Compare with the photospheric flux.

$\sigma T_{\rm eff}^4 = 6.3 \times 10^7$ W m^{-2}. Chromosphere: $4 \times 10^3 / 6.3 \times 10^7 \approx 6 \times 10^{-5}$; corona: $\approx 5 \times 10^{-6}$. Only a tiny fraction of the energy, about 10^{-4}, needs to be deposited non-radiatively (by waves or magnetic reconnection) to heat the outer atmosphere. The *temperature* is high because the density is so low that this small power must be radiated by very few particles.

Common misconceptions

1. **“The photosphere is the hottest visible layer.”** The temperature minimum lies above it, and the chromosphere and corona are much hotter, but they are faint in visible light because their density is low.
2. **“Line blanketing only removes light.”** The blocked energy is redistributed: it emerges in other frequency windows and heats the deeper layers (backwarming).
3. **“Model atmospheres need the stellar radius.”** For plane-parallel models only $T_{\rm eff}$, $\log g$ and composition matter; the radius enters only for extended (spherical) atmospheres.
4. **“1D LTE models are always good enough for abundances.”** For many elements, 3D and NLTE effects change abundances by 0.1–0.3 dex or more, especially in metal-poor stars and giants.

Lecture summary

- A model atmosphere solves hydrostatic equilibrium, energy conservation (radiation + convection), the RTE, the equation of state and the opacities self-consistently, as a function of $T_{\rm eff}$, $\log g$ and composition.
- Non-gray opacity causes blocking, backwarming and surface cooling. The dominant continuum opacity changes from electron scattering (O) to H I (A) to H$^-$ (F–K) to molecules (M).
- Convection appears where $\nabla_{\rm rad} > \nabla_{\rm ad}$; 3D radiation-hydrodynamics models granulation directly.

- The solar atmosphere: photosphere → temperature minimum (about 4200 K) → chromosphere → transition region → corona (1–2 MK), with non-radiative heating and NLTE above the minimum.
- Main codes and grids: ATLAS, MARCS, PHOENIX, TLUSTY, CMFGEN, and 3D codes (Stagger, CO5BOLD, MURaM, Bifrost).

Exam questions and answers — Lecture SA8

Q1. List the equations of a classical 1D LTE model atmosphere and explain why it is determined by $T_{\rm eff}$, $\log g$ and composition only.

> **A1.** Hydrostatic equilibrium $dP/d\tau = g/\kappa$ (plus radiation and turbulent pressure); energy conservation $F_{\rm rad} + F_{\rm conv} = \sigma T_{\rm eff}^4$; the RTE at many frequencies; mixing-length convection; the equation of state with Saha–Boltzmann populations; and opacities $\kappa_\nu(T, P, \text{composition})$. In plane-parallel geometry g is constant and the flux is fixed by $T_{\rm eff}$; the radius appears nowhere, so $T_{\rm eff}$, g and composition fully determine the structure.

Q2. What are line blanketing, backwarming and surface cooling?

> **A2.** Line blanketing is the blocking of flux by many lines, mainly in the UV. Because the total flux must still be carried, the radiation is redirected to other frequencies and partly back into the atmosphere, raising the temperature of the deeper layers (backwarming). In the upper layers, lines efficiently radiate energy away, lowering the boundary temperature below the gray value (surface cooling).

Q3. Describe the temperature structure of the solar atmosphere from the photosphere to the corona, and say which parts can be described by radiative equilibrium in LTE.

> **A3.** T falls from about 6400 K at $\tau_{500} = 1$ to a minimum of about 4200 K some 500 km higher (photosphere: radiative equilibrium, mostly LTE). It then rises through the chromosphere (6000–20 000 K), jumps in a thin transition region near 2100–2300 km, and reaches 1–2 MK in the corona. Only the photosphere is close to radiative equilibrium in LTE. The chromosphere and corona require non-radiative heating, and their lines must be treated in NLTE.

Q4. Using the pressure scale height, explain why red supergiants need spherical model atmospheres but the Sun does not.

> **A4.** $H = kT/(\mu m_H g)$. For the Sun $H \approx 134$ km, so the atmosphere (about $10H$) is 0.2 % of R: plane-parallel. For a red supergiant ($\log g \approx 0$, $R \approx 500 R_\odot$), $H \approx 3.6 R_\odot$ and $10H/R \approx 7\,\%$: rays leaving at different heights see different curvature, the outer layers dilute the radiation ($J \neq$ the plane-parallel value), and the luminosity depends on the radius at which it is defined. Spherical models are required.

Final Examination — Course B: Stellar Atmospheres (with complete solutions)

Format suggestion: 3 hours. Part A (20 %): short answers. Part B (50 %): answer any 6 of the 8 problems. Part C (15 %): one essay. Part D (15 %): classification. Solutions follow each question; numbers were checked with `scripts/lte_solvers.py` and `scripts/sa_physics.py`. Constants: $h = 6.626 \times 10^{-34}$ J s, $k = 1.381 \times 10^{-23}$ J K^{-1}, $c = 2.998 \times 10^{8}$ m s^{-1}, $\sigma = 5.670 \times 10^{-8}$ W m^{-2} K^{-4}, $m_e = 9.109 \times 10^{-31}$ kg, $m_H = 1.674 \times 10^{-27}$ kg, $r_e = 2.818 \times 10^{-15}$ m, $\sigma_T = 6.652 \times 10^{-29}$ m^2, $R_\odot = 6.957 \times 10^{8}$ m, 1 pc $= 3.086 \times 10^{16}$ m, 1 eV $= 1.602 \times 10^{-19}$ J. $\int_0^\infty E_n(t)\,dt = \dfrac{1}{n}$, $\int_0^\infty t\,E_n(t)\,dt = \dfrac{1}{n+1}$, $E_n(0) = \dfrac{1}{n-1}$.

Part A — Short answers (2 marks each)

A1. True or false: “In LTE the specific intensity equals the Planck function.”

> **False.** In LTE the **source function** equals $B_\nu(T)$. $I_\nu = B_\nu$ only in full TE, or deep inside an optically thick, isothermal region.

A2. What is the exact boundary temperature of a gray atmosphere in units of $T_{\rm eff}$?

> $(\sqrt{3}/4)^{1/4} = 0.811$.

A3. Why do Balmer lines peak in A stars?

> Boltzmann excitation into $n = 2$ increases with T, while Saha ionisation removes neutral H above about 10^4 K. The product peaks near 9900 K.

A4. Name the integral equation of the gray LTE atmosphere.

> Milne’s equation, $J(\tau) = \Lambda_\tau[J] = \tfrac{1}{2}\int_0^\infty J(t) E_1|t-\tau|dt$.

A5. Why are NLTE effects strongest in strong resonance lines?

> They form in high, low-density layers where $C \ll A$ ($\varepsilon \ll 1$), and the thermalisation depth $1/\sqrt{\varepsilon}$ is very

large.

A6. What does a positive β in $B = B_0(1 + \beta\tau)$ imply for limb darkening and for line formation in LTE?

Temperature increasing inward: limb darkening, and absorption lines.

A7. What is the threshold wavelength of H^- bound–free absorption, and why?

1.64 µm: the extra electron is bound by only 0.754 eV, and $\lambda = hc/0.754\,\mathrm{eV}$.

A8. Write the Klein–Rosseland relation between collisional excitation and de-excitation.

$q_{lu}/q_{ul} = (g_u/g_l)\, e^{-\Delta E/kT}$.

A9. In the coronal approximation, how does the upper-level population scale with electron density?

$n_u/n_l = n_e q_{lu}/A_{ul} \propto n_e$ (far below the Boltzmann value).

A10. Which part of a Voigt profile measures pressure, and what is its asymptotic form?

The far wings: $H(a, v) \approx a/(\sqrt{\pi} v^2)$, with $a \propto \gamma$, which grows with the perturber density.

Part B — Problems

B1. Kirchhoff's law and LTE (8 marks)

a. Using the Einstein coefficients, write j_ν and α_ν for a line. (b) Show that if the level populations obey Boltzmann's law at temperature T, then $S_\nu = B_\nu(T)$. (c) Explain why this result does not require $I_\nu = B_\nu$.

Solution. (a) $j_\nu = \frac{h\nu}{4\pi} n_2 A_{21} \phi$, $\alpha_\nu = \frac{h\nu}{4\pi}\phi(n_1 B_{12} - n_2 B_{21})$. (b) $S = \frac{n_2 A_{21}}{n_1 B_{12} - n_2 B_{21}} = \frac{A_{21}/B_{21}}{\frac{n_1 g_2}{n_2 g_1} - 1}$ (using $g_1 B_{12} = g_2 B_{21}$). Boltzmann gives $\frac{n_1 g_2}{n_2 g_1} = e^{h\nu/kT}$, and $A_{21}/B_{21} = 2h\nu^3/c^2$, so $S = B_\nu(T)$. (c) Only the **populations** (a property of the matter, set by collisions) were used. The radiation field does not appear, so $S = B$ holds whenever the matter is in LTE, whatever I_ν is.

B2. Saha equation (8 marks)

Pure hydrogen has $n_e = 10^{20}$ m^{-3}. Find the ionised fraction at (a) $T = 6000$ K and (b) $T = 8000$ K. ($U_\mathrm{I} = 2$, $U_\mathrm{II} = 1$, $\chi = 13.6$ eV.)

Solution. $\frac{N_{\rm II}}{N_{\rm I}} = \frac{1}{n_e}\left(\frac{2\pi m_e kT}{h^2}\right)^{3/2} e^{-\chi/kT}$ (the factor $2U_{\rm II}/U_{\rm I} = 1$). (a) $T = 6000$ K: $(\dots)^{3/2} = 2.415\times10^{21}\times 6000^{1.5} = 1.12\times10^{27}$ m^{-3}; $\chi/kT = 26.3$, so $e^{-26.3} = 3.8\times10^{-12}$. Ratio $= 1.12\times10^{27}\times3.8\times10^{-12}/10^{20} = 4.2\times10^{-5}$: essentially neutral. (b) $T = 8000$ K: $(\dots)^{3/2} = 1.73\times10^{27}$; $\chi/kT = 19.7$, so $e^{-19.7} = 2.7\times10^{-9}$. Ratio $= 0.047$; ionised fraction $= 0.045$. A 33 % rise in T increases ionisation by a factor of about 1000: Saha ionisation is extremely sensitive to temperature.

B3. The gray LTE atmosphere — full derivation (12 marks)

a. Starting from the plane-parallel RTE with $S = B$ and gray opacity, show that radiative equilibrium implies $J = B$ and H = constant. (b) Using the Eddington approximation, derive $T(\tau)$. (c) For $T_{\rm eff} = 10\,000$ K, tabulate T at $\tau = 0, 2/3, 1, 4$. (d) Derive the emergent intensity $I(0,\mu)$ and the limb-darkening law. (e) The exact surface temperature is $0.811\,T_{\rm eff}$. What is the percentage error of the Eddington value?

Solution. (a) RE: $\int \kappa_\nu (J_\nu - B_\nu)d\nu = 0$, which with $\kappa_\nu = \kappa$ gives $J = B$. The zeroth moment $dH/d\tau = J - S = J - B = 0$ gives H = const $= \sigma T_{\rm eff}^4/4\pi$. (b) $dK/d\tau = H \Rightarrow K = H\tau +$ const; $K = J/3 \Rightarrow J = 3H(\tau + C)$; $J(0) = 2H \Rightarrow C = 2/3$. $J = B = \sigma T^4/\pi$ gives $T^4 = \frac{3}{4}T_{\rm eff}^4(\tau + \frac{2}{3})$. (c) $T(0) = 8409$ K, $T(2/3) = 10\,000$ K, $T(1) = 10\,574$ K, $T(4) = 13\,678$ K. (d) $S = 3H(\tau + 2/3)$ is linear, so $I(0,\mu) = 3H(\mu + 2/3) = \frac{3\sigma T_{\rm eff}^4}{4\pi}(\mu + \frac{2}{3})$, and $I(\mu)/I(1) = (2 + 3\mu)/5$. (e) $(0.841 - 0.811)/0.811 = 3.7\,\%$.

B4. Milne–Eddington line in LTE (10 marks)

In an LTE atmosphere $B_\nu = B_0(1 + \beta\tau_c)$, and a line has a constant ratio $\eta_\nu = \eta_0 e^{-v^2}$ of line to continuum opacity. (a) Derive F_ν/F_c. (b) For $\beta = 3$, find the central depth for $\eta_0 = 0.1, 1, \infty$. (c) Show that for weak lines the equivalent width is proportional to η_0, and find the constant (in units of $\Delta\nu_D$).

Solution. (a) $\tau_\nu = (1 + \eta_\nu)\tau_c$, so B is linear in τ_ν and E–B is exact: $r = \frac{1 + 2\beta/[3(1+\eta)]}{1 + 2\beta/3}$. (b) $\beta = 3$: $1 + 2\beta/3 = 3$. $\eta_0 = 0.1$: $r = (1 + 2/1.1)/3 = 0.939$, depth 0.061. $\eta_0 = 1$: $r = 2/3$, depth 0.333. $\eta_0 \to \infty$: $r = 1/3$, depth 0.667. (c) For $\eta \ll 1$: $1 - r \approx \frac{2\beta/3}{1+2\beta/3}\eta_\nu$, so $W/\Delta\nu_D = \int (1 - r)dv = \frac{2\beta/3}{1+2\beta/3}\eta_0\sqrt{\pi}$. For $\beta = 3$: $W/\Delta\nu_D = \frac{2}{3}\sqrt{\pi}\,\eta_0 = 1.18\,\eta_0$.

B5. Hydrogen bound–free opacity and the Balmer jump (8 marks)

a. Compute Kramers' cross-section ($g_{\rm bf} = 1$) at the Balmer edge. (b) Give the wavelengths of the Paschen and Brackett edges. (c) Explain why the Balmer jump is large in A stars and small in the Sun.

Solution. (a) $\nu_2 = 3.288\times10^{15}/4 = 8.22\times10^{14}$ Hz; $\sigma = 2.815\times10^{29}/(2^5\nu_2^3)$ cm^2 $= 1.6\times10^{-17}$ cm^2 $= 1.6\times10^{-21}$ m^2. (b) $\lambda_n = 91.13\,n^2$ nm: Paschen 820 nm, Brackett 1458 nm. (c) Near 9000–10 000 K, hydrogen bound–free from $n = 2$ dominates the continuum; blueward of 364.7 nm the opacity jumps

by a factor of about 26, we see higher and cooler layers, and the flux drops. In the Sun $\mathrm{H^-}$ dominates by a factor of about 100 and has no edge at 364.7 nm, so the hydrogen jump is diluted.

B6. The negative hydrogen ion (8 marks)

Using Saha's equation with $\chi = 0.754$ eV, $U(\mathrm{H^-}) = 1$, $U(\mathrm{H\,I}) = 2$, compute $n(\mathrm{H^-})/n(\mathrm{H\,I})$ for $n_e = 10^{19}$ $\mathrm{m^{-3}}$ at (a) 6000 K and (b) 5000 K. Explain the trend and why metal-poor stars have less $\mathrm{H^-}$ opacity.

Solution. $n(\mathrm{H^-})/n(\mathrm{H\,I}) = n_e / \left[4(2\pi m_e kT/h^2)^{3/2} e^{-0.754\,\mathrm{eV}/kT}\right]$. (a) 6000 K: $(\dots)^{3/2} = 1.12 \times 10^{27}$ $\mathrm{m^{-3}}$, $e^{-1.458} = 0.233$, ratio $= 9.6 \times 10^{-9}$. (b) 5000 K: 8.5×10^{26}, $e^{-1.750} = 0.174$, ratio $= 1.7 \times 10^{-8}$. At fixed n_e, cooler gas holds more $\mathrm{H^-}$ (the weak binding is less easily broken). In a real atmosphere, however, n_e itself falls steeply with T. Since the electrons come from metals, metal-poor stars have lower n_e, hence less $\mathrm{H^-}$ and lower opacity.

B7. Transition rates and critical density (10 marks)

A forbidden line has $A_{ul} = 0.01$ $\mathrm{s^{-1}}$, $\Omega = 1$, $g_u = 3$, at $T = 8000$ K. (a) Compute q_{ul} and n_{crit}. (b) Find $(n_u/n_l)/(n_u/n_l)_{\mathrm{LTE}}$ at $n_e = 10^{10}$ $\mathrm{m^{-3}}$. (c) Show that in the coronal limit the ionisation balance is independent of density.

Solution. (a) $q_{ul} = 8.629 \times 10^{-12}\Omega/(g_u\sqrt{T}) = 8.629 \times 10^{-12}/(3 \times 89.4) = 3.2 \times 10^{-14}$ $\mathrm{m^3\ s^{-1}}$; $n_{\mathrm{crit}} = A/q_{ul} = 3.1 \times 10^{11}$ $\mathrm{m^{-3}}$. (b) With $x = n_e/n_{\mathrm{crit}} = 0.032$, the ratio is $x/(1+x) = 0.031$: the upper level holds 3 % of its LTE population. (c) Coronal limit: $n_i n_e q_{\mathrm{ion}}(T) = n_{i+1} n_e \alpha_{\mathrm{rec}}(T)$; n_e cancels, so $n_{i+1}/n_i = q_{\mathrm{ion}}/\alpha_{\mathrm{rec}}$ depends only on T.

B8. Broadening of the Na D line (8 marks)

For Na I D at 589 nm in the solar photosphere, take $T = 5772$ K, $m = 23$ u, $\xi = 1$ km $\mathrm{s^{-1}}$ and total damping $\gamma = 1.5 \times 10^9$ $\mathrm{s^{-1}}$. Compute (a) the Doppler width in mÅ, (b) the damping parameter a, (c) the distance from line centre (in mÅ) where the Lorentzian wing equals the Gaussian core.

Solution. (a) $v_D = \sqrt{2kT/m + \xi^2} = \sqrt{4.17 \times 10^6 + 10^6} = 2.27$ km $\mathrm{s^{-1}}$; $\Delta\lambda_D = \lambda v_D/c = 44.7$ mÅ. (b) $\Delta\nu_D = v_D/\lambda = 3.86 \times 10^9$ Hz, so $a = \gamma/(4\pi\Delta\nu_D) = 0.031$. (c) Solve $e^{-v^2} = a/(\sqrt{\pi}v^2)$: $v \approx 2.41$, i.e. $2.41 \times 44.7 \approx 108$ mÅ from centre. Beyond about 0.1 Å the wings are controlled by collisions, which is why the broad Na D wings measure pressure.

Part C — Essay (choose one, 10 marks)

C1. "Local thermodynamic equilibrium is a statement about matter, not about radiation." Discuss, with reference to Kirchhoff's law, the two-level atom and the thermalisation depth.

Model answer outline. (1) Define TE and LTE; Maxwell, Boltzmann and Saha at the local T. (2) Kirchhoff: $j = \alpha B$ depends only on the matter state, hence $S = B$ in LTE while $I \neq B$ (escape at the surface; $I(0, \mu < 0) = 0$; $J(0) \approx B/2$). (3) Validity: collisions must dominate radiative rates; two-level atom $S = (1-\varepsilon)\bar{J} + \varepsilon B$, so LTE when $\varepsilon \to 1$. (4) Even with ε close to but below 1, S deviates from B within $\tau \sim 1/\sqrt{\varepsilon}$ of the surface, with $S(0) = \sqrt{\varepsilon} B$. (5) Consequences: LTE is good for continua and weak lines in dwarf photospheres; it fails for strong lines, chromospheres, hot or low-density stars and nebulae. (6) Practical: LTE turns the RTE into an explicit formal solution once $T(\tau)$ is known; NLTE requires solving rate equations coupled to the radiation field (ALI).

C2. Describe how a model stellar atmosphere is computed, starting from the gray LTE solution, and how its predictions are compared with observations.

Model answer outline. Inputs $T_{\rm eff}$, $\log g$, composition → gray starting $T(\tau) = [\frac{3}{4} T_{\rm eff}^4 (\tau + q)]^{1/4}$ → hydrostatic $P(\tau)$ from $dP/d\tau = g/\kappa$ → Saha–Boltzmann populations → κ_ν, j_ν → solve the RTE at many frequencies (formal solution, or Feautrier/ALI with scattering) → check radiative equilibrium $\int \kappa_\nu (J_\nu - B_\nu) d\nu = 0$ and constant flux → temperature correction (Unsöld–Lucy) → iterate to convergence → emergent spectrum, limb darkening, line profiles (Milne–Eddington intuition, curve of growth) → compare with observed spectra, colours, Balmer jump and line equivalent widths → derive $T_{\rm eff}$, g and abundances.

Part D — Regimes and opacity (4 marks each)

D1. (Lecture SA7) Classify as TE, LTE or NLTE, with one sentence of justification each: (a) radiation 0.5 $R_\odot$ from the centre of the Sun; (b) the continuum at 500 nm in the solar photosphere; (c) the core of the Ca II K line; (d) [O III] 500.7 nm in a planetary nebula with $n_e = 10^{10}$ m^{-3}.

Solution. (a) **TE**: the photon mean free path (cm) is 10^{-10} of the temperature scale height, so the radiation is isotropic and Planckian at the local T. (b) **LTE**: H^- populations are set by collisions at $n_e \sim 10^{19}$–10^{20} m^{-3}, so $S = B$, although $I \neq B$ at the surface. (c) **NLTE**: a strong resonance line forming high in the low-density chromosphere, with photon losses ($S < B$) and non-radiative heating. (d) **NLTE**: $n_e \approx 0.016\, n_{\rm crit}$, so every collisional excitation is followed by radiative decay.

D2. (Lecture SA8) Explain line blanketing and backwarming, and name the dominant continuum opacity source in (a) an O star, (b) an A0 star, (c) the Sun, (d) an M dwarf.

Solution. Many lines block flux (mainly in the UV). The energy must still escape, so it is redistributed to other frequencies and partly returned to deeper layers, which become hotter (backwarming), while the outer layers cool. Opacity sources: (a) electron scattering and He bound-free; (b) H I bound-free (Balmer continuum); (c) H^- bound-free and free-free; (d) H^- plus molecules (TiO, H_2O).

Formula Sheet — Course B: Stellar Atmospheres

LTE (Lecture SA1)

- Kirchhoff: $j_\nu = \alpha_\nu B_\nu(T) \Rightarrow S_\nu = B_\nu(T)$ (but $I_\nu \neq B_\nu$)
- Boltzmann: $\dfrac{N_j}{N_i} = \dfrac{g_j}{g_i} e^{-\Delta E/kT}$ · Saha: $\dfrac{N_{r+1} n_e}{N_r} = \dfrac{2U_{r+1}}{U_r}\left(\dfrac{2\pi m_e kT}{h^2}\right)^{3/2} e^{-\chi/kT}$, $(2\pi m_e k/h^2)^{3/2} = 2.415\times10^{21}\ \mathrm{m^{-3}K^{-3/2}}$
- LTE opacity factor $1 - e^{-h\nu/kT}$ · two-level atom: $S = (1-\varepsilon)\bar{J} + \varepsilon B$, $\varepsilon' = \frac{C_{21}}{A_{21}}(1-e^{-h\nu/kT})$, $\varepsilon = \frac{\varepsilon'}{1+\varepsilon'}$ · $n_{\mathrm{crit}} = A_{21}/q_{21}$

Continuous opacity (Lecture SA2)

- Kramers: $\sigma_{\mathrm{bf}}(n,\nu) = 2.815\times10^{29} Z^4 g_{\mathrm{bf}}/(n^5\nu^3)\ \mathrm{cm^2}$ for $\nu \geq \nu_n = 3.288\times10^{15} Z^2/n^2$ Hz; edges $\lambda_n = 91.13\, n^2/Z^2$ nm
- LTE: $\alpha_\nu^{\mathrm{bf}} = \sum_n N_n \sigma_{\mathrm{bf}}(n,\nu)(1-e^{-h\nu/kT})$, $N_n = N_{\mathrm{H\,I}} 2n^2 e^{-E_n/kT}/U$
- Free–free: $\alpha_\nu^{\mathrm{ff}} = 3.69\times10^8 Z^2 n_e n_i T^{-1/2}\nu^{-3} g_{\mathrm{ff}}(1-e^{-h\nu/kT})\ \mathrm{cm^{-1}}$ (cgs)
- $\mathrm{H^-}$: $\chi = 0.754$ eV, threshold 1.64 µm, $\sigma_{\max} \approx 4\times10^{-17}\ \mathrm{cm^2}$ near 850 nm; $\frac{n(\mathrm{H^-})}{n(\mathrm{H\,I})n_e} = \frac{1}{4}\left(\frac{h^2}{2\pi m_e kT}\right)^{3/2} e^{0.754\,\mathrm{eV}/kT}$
- Thomson: $\sigma_T = 6.652\times10^{-29}\ \mathrm{m^2}$, $\kappa_{\mathrm{es}} = 0.020(1+X)\ \mathrm{m^2/kg}$ · Rayleigh $\propto \lambda^{-4}$ · Kramers' law $\bar{\kappa} \propto \rho T^{-3.5}$

Transition rates (Lecture SA3)

- Statistical equilibrium: $n_i \sum_j (R_{ij} + C_{ij}) = \sum_j n_j (R_{ji} + C_{ji})$
- Photo-ionisation: $R_{i\kappa} = 4\pi \int \frac{\sigma_i}{h\nu} J_\nu d\nu$ · recombination (Milne): $R_{\kappa i} = \left(\frac{n_i}{n_\kappa}\right)^* 4\pi \int \frac{\sigma_i}{h\nu}\left(\frac{2h\nu^3}{c^2} + J_\nu\right) e^{-h\nu/kT} d\nu$
- Collisions: $q_{ul} = 8.629\times10^{-6}\Omega/(g_u\sqrt{T})\ \mathrm{cm^3\ s^{-1}}$ · Klein–Rosseland: $q_{lu} = (g_u/g_l) q_{ul} e^{-\Delta E/kT}$
- Critical density $n_{\mathrm{crit}} = A_{ul}/q_{ul}$ · two-level, no radiation: $\frac{(n_u/n_l)}{(n_u/n_l)^*} = \frac{n_e/n_{\mathrm{crit}}}{1+n_e/n_{\mathrm{crit}}}$
- Coronal: $n_u/n_l = n_e q_{lu}/A_{ul}$; $n_{i+1}/n_i = q_{\mathrm{ion}}(T)/\alpha_{\mathrm{rec}}(T)$ · Nebular: $J_\nu \approx W B_\nu(T_*)$, $W \approx \frac{1}{4}(R_*/r)^2$

Line absorption and broadening (Lecture SA4)

- $\alpha_\nu^{\mathrm{line}} = \frac{\pi e^2}{m_e c} f_{lu} n_l \phi(\nu)(1-e^{-h\nu/kT})$, $\frac{\pi e^2}{m_e c} = 0.02654\ \mathrm{cm^2\ Hz}$ · $A_{ul} = 0.6670 \frac{g_l}{g_u}\frac{f_{lu}}{\lambda^2[\mathrm{cm}]}\ \mathrm{s^{-1}}$
- Natural: $\gamma_{\mathrm{rad}} = \Gamma_u + \Gamma_l$, $\Delta\lambda_{\mathrm{FWHM}} = \lambda^2\gamma/(2\pi c)$; classical $\gamma_{\mathrm{cl}} = 2.47\times10^{-22}\nu^2\ \mathrm{s^{-1}}$
- Doppler: $\Delta\nu_D = \frac{\nu_0}{c}\sqrt{2kT/m + \xi^2}$ · pressure: $\Delta\nu = C_n/r^n$, $n = 2$ (linear Stark), 3 (resonance), 4 (quadratic Stark), 6 (van der Waals); impact width $\gamma_n \propto N C_n^{2/(n-1)} \bar{v}^{(n-3)/(n-1)}$
- Voigt: $\phi = H(a,v)/(\sqrt{\pi}\Delta\nu_D)$, $a = \gamma/(4\pi\Delta\nu_D)$, wings $H \approx a/(\sqrt{\pi}v^2)$ · rotation: $\Delta\lambda_L = \lambda v \sin i/c$

Gray LTE atmosphere (Lecture SA5)

- $J = B = S$, $H = \sigma T_{\rm eff}^4/4\pi$
- Eddington: $T^4 = \frac{3}{4}T_{\rm eff}^4(\tau + \frac{2}{3})$, $T(0) = 0.841 T_{\rm eff}$, $T(2/3) = T_{\rm eff}$
- Exact: $T^4 = \frac{3}{4}T_{\rm eff}^4[\tau + q(\tau)]$, $q(0) = 1/\sqrt{3}$, $q(\infty) = 0.7104$, $T(0) = 0.811 T_{\rm eff}$
- Limb darkening: Eddington $I(\mu)/I(1) = (2 + 3\mu)/5$; exact $H(\mu)/H(1)$, $I(0)/I(1) = 0.344$
- Two-stream ($\mu = \pm 1/\sqrt{3}$): $T^4 = \frac{3}{4}T_{\rm eff}^4(\tau + 1/\sqrt{3})$
- Hydrostatic: $dP/d\tau = g/\kappa \Rightarrow P = g\tau/\kappa$

Lines (Lecture SA6)

- $\int \sigma_\nu d\nu = \pi r_e c f \cdot \Delta\nu_D = \frac{\nu_0}{c}\sqrt{2kT/m + \xi^2}$ · Voigt $a = \gamma/4\pi\Delta\nu_D$
- Milne–Eddington (LTE): $\dfrac{F_\nu}{F_c} = \dfrac{1 + 2\beta/[3(1+\eta_\nu)]}{1 + 2\beta/3}$, maximum depth $1 - \frac{1}{1+2\beta/3}$
- Weak line: $W_\lambda = \pi r_e \lambda^2 N f$ · Curve of growth: $W \propto N \to \sqrt{\ln N} \to \sqrt{N}$
- NLTE: $S^{\rm line} = \frac{2h\nu^3}{c^2}\left[\frac{b_l}{b_u}e^{h\nu/kT} - 1\right]^{-1}$

TE, LTE, NLTE (Lecture SA7)

- TE: $I_\nu = B_\nu(T)$ · LTE: $S_\nu = B_\nu(T(\mathbf{r}))$ · NLTE: rate equations, $S_\nu \neq B_\nu$
- Statistical equilibrium: $n_i \sum_j (R_{ij} + C_{ij}) = \sum_j n_j(R_{ji} + C_{ji}) \cdot b_i = n_i/n_i^{\rm LTE}$
- $S^{\rm line} = \frac{2h\nu^3}{c^2}\left[\frac{b_l}{b_u}e^{h\nu/kT} - 1\right]^{-1} \approx \frac{b_u}{b_l}B_\nu$ · criterion: n_e vs $n_{\rm crit} = A\beta_{\rm esc}/q$

Stellar atmospheres (Lecture SA8)

- $dP/d\tau = g/\kappa \cdot F_{\rm rad} + F_{\rm conv} = \sigma T_{\rm eff}^4$ · scale height $H = kT/(\mu m_H g)$
- Convection if $\nabla_{\rm rad} > \nabla_{\rm ad}$ (0.4 ideal monatomic); gray atmosphere: $\nabla_{\rm rad} = \tau/[4(\tau + 2/3)] < 0.25$